Numerical Modeling of Two-Phase Condensation in a Microchannel Condenser Using R245fa

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1. Introduction

Efficient thermal management is critical in high-power electronic systems, where large heat fluxes must be dissipated within compact geometries. Microchannel condensers are widely used in such applications due to their high surface-area-to-volume ratio and ability to sustain efficient phase-change heat transfer.

In this project, the performance of a microchannel condenser using R245fa as the working fluid is analyzed. The refrigerant enters the condenser as saturated or near-saturated vapor and undergoes condensation as heat is rejected to an external air stream. The objective is to model this process by resolving the coupled heat transfer and fluid flow behavior along the condenser.

The analysis is performed using a segment-by-segment approach, in which the condenser is discretized into small axial elements. For each segment, local thermodynamic properties, heat transfer rates, vapor quality, void fraction, and pressure drop are evaluated. This allows the progressive condensation process and associated changes in flow behavior to be captured.

Key physical models employed include:

- Lagrange interpolation for thermophysical properties
- The Traviss correlation for condensation heat transfer
- The Lockhart–Martinelli relation for void fraction
- The Friedel correlation for two-phase pressure drop

In Task 1, a single condenser segment is analyzed to establish the modeling framework and verify the implementation of the governing equations.

In Task 2, this model is extended to the full condenser by applying the segment calculation iteratively along the flow direction. The outlet conditions of each segment serve as the inlet conditions for the next, allowing the evolution of pressure, vapor quality, and void fraction to be tracked throughout the condenser. This enables evaluation of overall condenser performance and the total extent of condensation.

2. Task 1 – Single Segment Condenser Analysis

2.1 Problem Description

A single segment of a microchannel condenser is analyzed to determine the thermodynamic state of R245fa after partial condensation. The objective is to compute the outlet vapor quality x_{out}, void fraction α_{out}, and pressure P_{out} over a segment of length $\Delta z = 0.04$ m.

The refrigerant enters the segment as nearly saturated vapor at a known pressure, while heat is rejected to air flowing across the condenser.

2.2 Given Conditions

The operating conditions for the segment are:

- Heat load: $\dot{q}_{chip}$ = 340 W
- Air inlet temperature: $T_{a,in}$ = 20°C
- Air velocity: V_a =6.5 m/s
- Refrigerant inlet pressure: P_{in} = 220 kPa
- Inlet vapor quality: x_{in} = 0.999
- Tube diameter: D = 3 mm
- Number of parallel passages: n = 9
- Segment length: Δz = 0.04 m
- Air properties are assumed constant:
    - ρ =1.3 kg/m3
    - c_p=1005 J/kgK

2.3 Thermophysical Properties

Saturation properties of R245fa at the inlet pressure are determined using the correlation:

$$T_{sat} = \frac{C_1}{ln(P) - C_0}$$

with C_0 = 20.472 and C_1 = −2579.9, where pressure is in Pa.

All remaining thermophysical properties (ρ_l, ρ_v, μ_l, μ_v, k_l, h_{lv}, σ, c_p) are obtained via Lagrange interpolation using tabulated saturation data over the temperature range 15–60°C *(Appendix A1)*. The interpolated properties were used as input to the segment model in Task 1. In particular, the saturation temperature defines the refrigerant temperature within the condensing segment, while the remaining properties are required for the condensation heat transfer coefficient, void fraction, and pressure drop calculations.

2.4 Methodology

2.4.1 Mass Flux

The refrigerant mass flux is computed from the imposed heat load and latent heat:

$$G = \frac{\dot{q}_{chip}}{nAh_{lv}}$$

where $A = \frac{\pi D^2}{4}$ is the cross-sectional area of a single passage. The mass flux G represents the amount of refrigerant flowing per unit area. It links the heat load to the required phase change rate: a higher heat load requires a higher mass flow rate to sustain condensation.

2.4.2 Condensation Heat Transfer Coefficient

The refrigerant-side heat transfer coefficient h_r is calculated using the Traviss correlation. This requires evaluation of:

- Liquid Reynolds number Re_l
- Martinelli parameter X_{tt}
- Prandtl number Pr_l

The coefficient h_r represents the effectiveness of heat transfer from the condensing refrigerant to the channel wall. It depends on both the liquid film behavior and the interaction between liquid and vapor phases. Higher values of h_r indicate more efficient condensation heat transfer.

2.4.3 Void Fraction

The vapor void fraction is calculated using the Lockhart–Martinelli relation:

$$\alpha = \left[1 + \left(\frac{1-x}{x}\right)^{0.64} \left(\frac{\rho_v}{\rho_l}\right)^{0.36} \left(\frac{\mu_l}{\mu_v}\right)^{0.07}\right]^{-1}$$

This is evaluated at both inlet and outlet conditions. The void fraction α represents the fraction of the channel volume occupied by vapor. A value close to 1 indicates vapor-dominated flow, while lower values indicate increasing liquid presence. As condensation progresses, α decreases, reflecting the formation of liquid within the flow.

2.4.4 Segment Heat Transfer

The total thermal conductance of the segment is:

$$UA_{seg} = \left(\frac{1}{h_r A_r} + \frac{1}{UA_{air}}\right)^{-1}$$

where:

- $A_r = n\pi D\Delta z$ is the refrigerant-side area
- UA_{air} is given from an empirical air-side relation

This expression represents two thermal resistances in series: one associated with condensation inside the tubes and one associated with heat transfer to the air. The overall conductance UA_{seg} quantifies how easily heat can pass from the refrigerant to the air stream.

To determine how much heat is actually transferred, the conductance must be compared to the ability of the air to absorb heat. This is captured through the number of transfer units (NTU):

$$NTU = \frac{UA_{seg}}{C_a}$$

where $C_a = \dot{m}_a c_p$ is the heat capacity rate of the air. A higher NTU indicates that heat transfer is strong relative to the air's ability to absorb energy. The segment heat transfer rate is then computed using an effectiveness–NTU formulation:

$$\dot{q}_{seg} = C_a(T_{sat} - T_{a,in})\left(1 - e^{-NTU}\right)$$

This expression accounts for the fact that the air temperature increases as it absorbs heat, which limits the maximum achievable heat transfer. The term $(1 - e^{-NTU})$ represents the effectiveness of the segment, i.e., the fraction of the maximum possible heat transfer that is actually achieved.

2.4.5 Exit Quality

The outlet vapor quality is obtained from an energy balance:

$$x_{out} = x_{in} - \frac{\dot{q}_{seg}}{\dot{q}_{chip}}$$

A lower bound is imposed to prevent non-physical negative quality. The vapor quality x represents the mass fraction of vapor in the refrigerant. A decrease in x indicates condensation. The change in quality is directly proportional to the heat removed in the segment.

2.4.6 Pressure Drop

The pressure drop is evaluated using a separated flow model with the Friedel correlation:

$$\phi_{lo}^2 = E + \frac{3.24FH}{Fr^{0.045}We^{0.035}}$$

The outlet pressure is then computed as:

$$P_{out} = P_{in} - \phi_{lo}^2 \left(\frac{dP}{dz}\right)_{lo} \Delta z - (\Gamma_{out} - \Gamma_{in})$$

The pressure drop arises from both frictional and acceleration effects in the two-phase flow. The Friedel correlation accounts for the complex interaction between liquid and vapor phases by scaling the liquid-only pressure gradient with a two-phase multiplier ϕ_{lo}^2.

The terms E, F, and H capture the influence of phase distribution, fluid properties, and flow regime, while the Froude and Weber numbers account for the effects of gravity and surface tension.

The acceleration term (Γout−Γin) reflects changes in momentum due to variations in vapor quality and phase distribution along the segment.

2.5 Results

A summary of the computed results for the segment can be found in *Table 1*, and the Python code can be found in *Appendix A2*.

Table 1 | *Summary of results Single Segment Condenser Element*

Quantity	Value	Unit
Saturation temperature $T_{sat,\,in}$	42.60	°C
Mass flux G	29.80	$kg \cdot m^{-2} \cdot s^{-1}$
Heat transfer coefficient h_r	2731.50	$W \cdot m^{-2} \cdot K^{-1}$
Segment conductance UA_{seg}	2.314	$W \cdot K^{-1}$
Segment heat transfer	38.50	W
Outlet vapor quality x_{out}	0.8858	-
Inlet void fraction α_{in}	0.9969	-
Outlet void fraction α_{out}	0.9347	-
Two-phase multiplier ϕ_{lo}^2	26.06	-
Outlet pressure P_{out}	219.98	kPa

2.6 Discussion

The results indicate that a measurable degree of condensation occurs within a single segment, as reflected by the decrease in vapor quality from x_{in} = 0.999 to x_{out} ≈ 0.886. This change confirms that the segment removes a non-negligible portion of the latent heat, despite its short length. However, the relatively high outlet quality shows that the refrigerant remains predominantly in the vapor phase, and that full condensation requires multiple segments.

The corresponding decrease in void fraction from α_{in} ≈ 0.997 to α_{out} ≈ 0.935 reflects the formation of liquid within the flow. While the void fraction remains high, indicating vapor-dominated flow, the reduction is consistent with the onset of condensation and increased liquid holdup. This change is important because it directly influences both heat transfer and pressure drop in subsequent segments.

The segment heat transfer rate ($\dot{q}_{seg}$ ≈ 38.5 W) is governed by the balance between thermal conductance and air-side capacity. The NTU value (≈0.65) corresponds to an effectiveness of approximately 0.48, meaning that only about half of the maximum possible heat transfer is achieved.

This indicates that the air-side heat capacity limits the heat transfer process, rather than the refrigerant-side resistance alone. In other words, even though the condensation heat transfer coefficient is relatively high, the air stream cannot absorb all the available heat in a single segment.

The pressure drop across the segment is small ($\Delta P \approx 0.017$ kPa), which is expected given the short segment length and moderate mass flux. The two-phase multiplier $\phi_{lo}^2 \approx 26$ indicates that two-phase effects significantly increase the frictional pressure drop compared to a liquid-only flow. However, because the base (liquid-only) pressure gradient is itself small, the overall pressure drop remains limited.

Additionally, the acceleration term contributes positively to the pressure balance due to the decrease in vapor quality, partially offsetting the frictional losses. This highlights that, in condensing flows, both frictional and momentum effects must be considered to accurately capture pressure evolution.

Overall, the results show that heat transfer dominates over pressure losses in a single segment. The segment effectively reduces vapor quality, but multiple segments are required to achieve complete condensation. This justifies the need for the segment-by-segment approach used in Task 2, where cumulative effects along the condenser length become significant.

3. Task 2 – Combined Evaporator and Condenser System Performance

3.1 Problem Description

The performance of the entire system with multiple segments is to be analyzed in Part 1 of Task 2. Starting from the interface between the chip and the evaporator, we analyze the system at five different evaporator pressures: P2 = 160, 180, 200, 220 and 240 kPa. Then, for each of these pressures, we must find the saturation properties of R-245fa, including the saturation temperature at the inlet of the first segment of the condenser tube, which is where the vapor from the evaporator in contact with the chip surface enters the condenser. Assuming negligible thermal resistance at the evaporator wall allows us to assume $T_{chip} = T_w$ and we are asked to find T_w. We are asked to plot the temperature of the chip versus the evaporator pressure and use this data to determine the evaporation pressure that will deliver the specified $\dot{q}_{chip}$ at the design chip temperature limit of T_{chip}= 55 degrees C.

In Part 2, we are asked to construct a loop to sequentially analyze the 20 segments of the condenser. The quality, pressure, and void fraction of each segment's outlet are to be found and utilized as the inlet conditions of the next segment of the condenser. Note that the effects of the return bends are neglected, they are treated as segments with no pressure drop or heat transfer. The final objectives in Part 2(a) are to determine the total heat transfer rate in the condenser, the total refrigerant pressure drop, and the exit state of the R-245fa. As well, in Part 2(b), we are asked to find the total condenser tube length value that would result in an exact quality of zero indicating only saturated liquid at the condenser outlet.

3.2 Given Conditions

The operating conditions for the segment are:

- Heat load: $\dot{q}_{chip}$ = 340 W
- Air inlet temperature: $T_{a,in}$ = 20°C
- Air velocity: V_a = 6.5 m/s
- Length of each segment: Δz = 0.04 m
- Number of segments (iterations of the loop): i = 20
- Tube (passage) diameter: D = 3 mm = 0.003 m
- Number of parallel passages: n = 9
- $\theta = 35\ degrees$
- Design Limit of T_{chip}= 55 °C
- Area of chip: $A_{chip} = 6\ cm \times 6\ cm$ = 0.0036 m
- $q_{double\ dot} = \dot{q}_{chip} / A_{chip} = 340\ W/0.0036m$ = 94.44 kW/m
- Inlet vapor quality: x_{s0}= 0.99fx9
- Inlet void fraction: $\alpha_{s0} = 0.999$
- As well in order to avoid a negative computed exit quality, we will set:
    - $\dot{q}_{seg} = 0.00001$ at the exit and set the exit quality to $x_{out} = 0.00001$.

- Air properties are assumed constant:
  - ρ =1.3 kg/m3
  - c_p=1005 J/kgK

3.3 Thermophysical Properties

Saturation properties of R245fa at the inlet are determined using the correlation:

$$T_{sat} = \frac{C_1}{ln(P_2) - C_0}$$

where P_2 (evaporator pressure) is to be determined by the program, C_0 = 20.472, and C_1 = −2579.9, where pressure is in Pa.

Then, the saturation properties including T_{sat} can be used to find $T_{chip} = T_w$ using the nucleate boiling correlation:

$$\frac{q_{double\ dot} D_b}{k_l(T_w - T_{sat})} = 207\left(\frac{q_{double\ dot} D_b}{k_l T_{sat}}\right)^{0.745}\left(\frac{\rho_v}{\rho_l}\right)^{0.581}(Pr_l)^{0.533}$$

$$D_b = 0.0208\theta\sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}}\ ,\quad \theta = 35\ deg\ \mathrm{C}$$

In conclusion, the following equation is derived and used in the program to find T_w:

let B = $\left(\frac{\rho_v}{\rho_l}\right)^{0.581}(Pr_l)^{0.533}$, then:

$$T_w = \left[\frac{1}{207}\left(\frac{q_{double\ dot} D_b}{k_l}\right)\left(\frac{k_l T_{sat}}{q_{double\ dot} D_b}\right)^{0.745} B^{-1}\right] + T_{sat}$$

All remaining thermophysical properties (ρ_l, ρ_v, μ_l, μ_v, k_l, h_{lv}, σ, c_p) are obtained via Lagrange interpolation using tabulated saturation data over the temperature range 15–60°C *(Appendix A1)*.

3.4 Methodology

The same thermophysical properties used in Task 1 such as void fraction, exit quality, pressure drop, mass flux, condensation heat transfer coefficient, and segment heat transfer were employed in this section. The code developed in Project 1 and subsequently Task 1 above was used to find the saturation properties of R245fa and the saturation temperature at the inlet of each section. This is carried out as the first step of the loop that will simulate the condensation of vapour in all segments of the condenser.

For Part 1, we utilized the Lagrange interpolation code developed in Task 1 to find the saturation values of the required properties to solve for T_{chip} via finding T_{sat} at each of the five evaporator pressures provided. As a result we found the relationship between the chip temperature and the evaporator pressure (see Figure 1 below). This loop was implemented in a function called

"T_wall_calculator" which took the chip rate of heat transfer, area of the chip, the design temperature limit, and the pressure values as inputs and provided a list consisting of the chip temperature (wall temperature) at each pressure value called "T_chip_list". Please refer to Appendix B below to see the full code.

Next, we fit a degree 1 polynomial to the data in Figure 1, to find the pressure that would be required to achieve our design requirements at the temperature limit of 55 °C (See Figure 2 below and see the code in Appendix B). This design evaporator pressure value called P_design in the code, will be carried forward and used in the iterative simulation to analyze the performance of the total length of the condenser in Part 2 of Task 2.

For Part 2(a), the thermophysical properties found for the refrigerant using the Lagrange method were used in addition to the design pressure to set up a loop for the 20 segments of the condenser. The function is called "condenser_sim" in the program and it houses a for loop that solves the pressure drop and quality value in addition to the heat transfer and some other properties of each segment in an iterative manner for the entirety of the condenser tube. The function outputs and results can be seen in the Results section below. We note that the total pressure loss and the total heat transfer rate values of the condenser are equal to the sum of each property's values for all individual segments in the condenser.

Then for Part 2(b), using the program we experimented with different total condenser tube lengths in order to predict the total condenser tube length that would result in an exact quality of zero indicating only saturated liquid at the condenser outlet. For this experimentation, a loop was used that searched the segments along the length of the condenser for their x_{outlet} value and pressure drop until the x_{out} dropped below the critical point of 0.0001 at which point it would be considered equal to zero. This loop was housed inside a function that could be called to provide the requested final length value. The function is called "best_length_search". Note that the condenser tube length was incrementally increased by the length of one segment at each iteration. The findings are reported and discussed in the Results and Discussion sections below.

3.5 Results

3.5.1 Task 2 Part 1 - Relation between Chip Temperature and the Evaporator Pressure

Figure 1 below shows the plot of temperature of the chip versus the evaporator pressure. A curve-fit of degree 1 polynomial was used to determine the evaporation pressure that will deliver the specified $\dot{q}_{chip}$ at the design chip temperature limit of T_{chip}= 55 °C. The design evaporation pressure was found as **285.63 kPa** (see Figure 2 below).

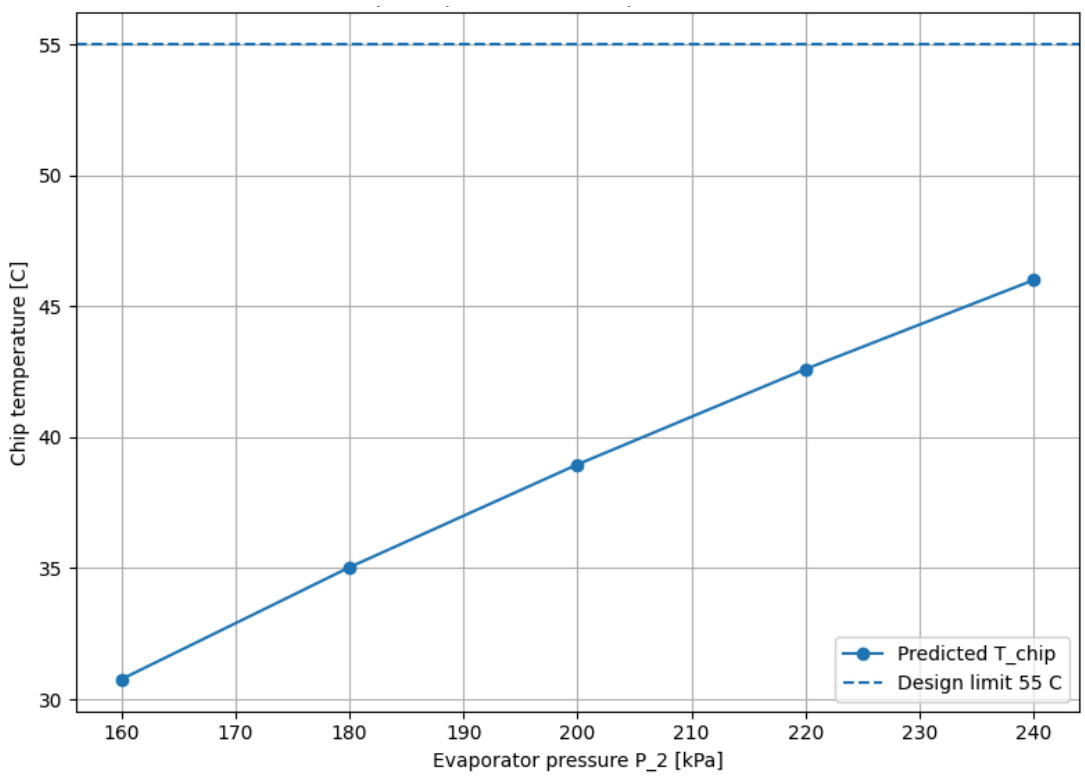

Figure 1. Chip Temperature versus the Evaporator Pressure Plot

```
Design evaporator pressure: 285.63 kPa, chip will be at 55C
```

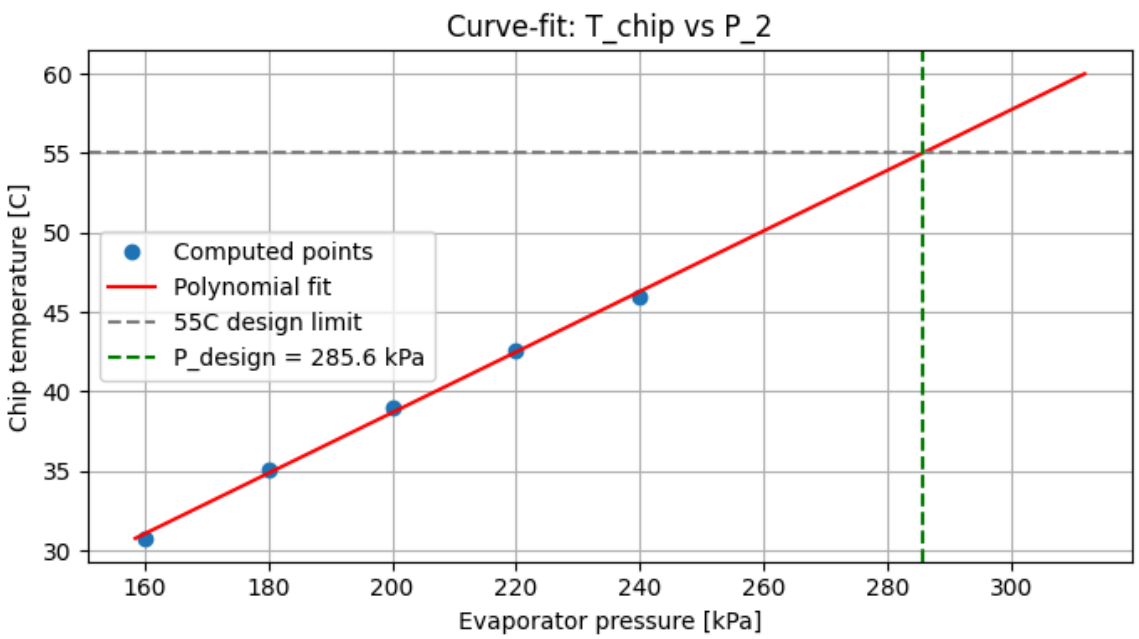

Figure 2. Program output for the fitted pressure at which T_chip = 55 °C: found to be **285.63 kPa**.

3.5.2 Task 2 Part 2 - Full Length Condenser Loop Results

As requested in Part 2(a), a summary of the computed results for the full condenser comprising all its segments, can be found below in *Table 2* and Figure 3. As well, the Python code that yielded the following results can be found in Appendix B.

Table 2 | *Summary of Cumulative Properties for the Full Condenser Length*

Cumulative Properties	**Value**	**Unit**
Total Rate of Heat Transfer from the Condenser to the Air (q_dot)	339.66	W
Total Refrigerant Pressure Drop (ΔP)	0.1228	kPa
Exit State of R-245fa out of the Condenser: Saturated Liquid (Fully Condensed) based on the properties reported below at the exit.		
Exit: Outlet vapor quality x_{out}	0.000010	-
Exit: Outlet void fraction α_{out}	0.002186	-
Exit: Outlet pressure P_{out}	285.5089	kPa

Seg	x_out	α_out	P_out (kPa)	q_seg (W)
1	0.83168	0.90611	285.6209	56.8901
2	0.64013	0.83386	285.5943	65.1266
3	0.44096	0.74891	285.5687	67.7177
4	0.24164	0.62545	285.5465	67.7671
5	0.04256	0.32129	285.5299	67.6888
6	0.00001	0.00219	285.5231	14.4663
7	0.00001	0.00219	285.5220	0.0000
8	0.00001	0.00219	285.5210	0.0000
9	0.00001	0.00219	285.5200	0.0000
10	0.00001	0.00219	285.5190	0.0000
11	0.00001	0.00219	285.5180	0.0000
12	0.00001	0.00219	285.5170	0.0000
13	0.00001	0.00219	285.5160	0.0000
14	0.00001	0.00219	285.5149	0.0000
15	0.00001	0.00219	285.5139	0.0000
16	0.00001	0.00219	285.5129	0.0000
17	0.00001	0.00219	285.5119	0.0000
18	0.00001	0.00219	285.5109	0.0000
19	0.00001	0.00219	285.5099	0.0000
20	0.00001	0.00219	285.5089	0.0000

Figure 3. Summary of Results for the Full Condenser Length - Quality (x), Void Fraction (α), Pressure at the Outlet of each Segment and the Rate of Heat Transfer to the Air

As requested in Part 2(b), we determined the total condenser tube length value that would result in an exact quality of zero indicating only saturated liquid at the condenser outlet. It was found that the total length of the condenser that results in a final exit quality of zero is **0.240 m.**

Code output:

```
[Part 2b] The total condenser length that achieves a quality of zero at the exit is: 0.240 m
```

3.6 Discussion

In Figure 1 above, we see that with the increase in chip temperature, the evaporator pressure increases. This makes sense because by increasing the temperature, movement and the average velocity of the refrigerant molecules increases in their rigid evaporator container, and they collide more frequently with the container walls resulting in a higher saturation pressure. Next, in Figure 2, we can see the pressure at which the design limit temperature of 55 °C is achieved was found to be 285.63 kPa. This agrees with the existing trend in the graph and where it intersects the 55 °C temperature constant horizontal line.

In Part 2, we found the properties at the outlet of each of the 20 segments of the condenser tube, in addition to the total rate of heat transfer, pressure drop, and the final state at the exit of the condenser tube. Analyzing these results reported in Table 2 and Table 3 above, we can conclude that the refrigerant condenses continuously while travelling through the condenser segments. Since its vapor quality and void fraction values decrease from the inlet values of 0.999 to almost zero: 0.0001 for the exit quality and 0.00219 for the void fraction at the outlet of segment 20. This reduction in the vapor quality is expected as this is a condenser, and inside of it, the refrigerant progressively rejects heat to the air around the condenser. Gradually decreasing the vapor fraction after each segment is travelled by the refrigerant along the tube, until a fully condensed saturated liquid is achieved at the exit of the tube. The pressure value at the output of each segment experiences a small gradual decrease along the tube as well. This happens due to frictional forces (discussed in Task 1) which result in a total pressure loss of 0.1228 kPa for the refrigerant. Note that compared to the condenser pressure of 285.6 kPa, this is so small that it can be neglected if one seeks to carry out some quick analysis for designing such a system. So, we can conclude that the saturation temperature and fluid properties are not impacted by the pressure drop in the condenser.

Note that the input heat transfer rate to the program was 340 W, and the total rate of heat transfer to the air around the condenser was found as 339.66 W which is very close to 340 W. The small error of 0.34W might be due to the correction made by the program in the calculation of the properties of the final segment by using a value of 0.00001 for the exit quality instead of using an impossible negative quality value. This would impact the final heat transfer value calculated for that segment. Then the total heat transfer rate of the condenser can be found by summing up all of the segments' heat transfer rates. This small error does not significantly affect the results and the energy balance in the system so it can be neglected and the results are still valid.

Finally, in Part 2(b) of this task, we found the total condenser length of 0.240 m corresponds to an exact vapor quality of zero at the exit of the condenser tube, which would indicate that there is only saturated liquid exiting the condenser. This tells us that the condenser tube must be at minimum 0.24 m long. If each segment is 0.04m, then that would mean 6 segments are required to achieve full condensation under the specified operating conditions for a system with a R-245fa working fluid

(refrigerant). A shorter tube would not achieve full condensation, so the state of the refrigerant at the exit would be a two-phase mixture. We know that two-phase mixtures can decrease the efficiency of the system by increasing the pressure drop and adding more complexity to the cooling loop. Given the fact that we have analyzed a very long condenser consisting of 20 segments in this exercise, and have ensured that it can achieve full condensation at the exit, under the specific design operating conditions, we can have confidence in the robustness and practicality of the system. That is to say, if there is a need to modify the surrounding temperature or chip temperature, our system is robust enough to be able to assuredly cool the chip by adding 14 more segments (for a total of 20 segments) and continue to perform as we expect.

4. Task 3 –Evaluation and Optimization of an Alternate Condenser Tube Design

In Task 3, the objective is to assess the impact of a proposed design modification on condenser performance. The new design replaces the original geometry with a flat tube containing ten circular refrigerant passages, each with an inner diameter of 2.4 mm. This modification alters the total flow area, hydraulic diameter, and heat transfer surface, and therefore affects both the condensation heat transfer and the two-phase pressure drop.

4.1 Geometry Adjustment

The goal of this section is to implement this new geometry in the existing model and compare its performance with the original design under identical operating conditions.

4.1.1 Methodology

To evaluate the modified condenser design, the code developed in Task 2 was adapted to reflect the new tube geometry. Specifically, the number of refrigerant passages was increased from 9 to 10, and the passage diameter was reduced from 3.0 mm to 2.4 mm.

All other operating conditions were kept identical to those used in Task 2, including:

- Heat dissipation rate: 340 W
- Inlet air temperature: 20 °C
- Air velocity: 6.5 m/s
- Segment length: 0.040 m (20 segments total)
- Evaporator pressure: same as determined in Task 2 (i.e., 285.63 kPa)

The condenser was modeled as a sequence of 20 segments. For each segment, the following quantities were computed:

- Exit vapor quality
- Exit void fraction
- Exit pressure
- Segment heat transfer rate

Saturated refrigerant properties were evaluated at the inlet pressure of each segment, consistent with the approach used in Task 2.

The simulation was performed for both the original design (9 × 3.0 mm) and the modified design (10 × 2.4 mm). The results were compared on a segment-by-segment basis, and overall condenser performance metrics (total heat transfer and total pressure drop) were also evaluated.

The complete Python code can be found in *Appendix C1*.

4.1.1 Results & discussion

The vapor quality decreases monotonically along the condenser length for both configurations, reaching approximately zero at segment 6, indicating complete condensation of the refrigerant *(Fig. 4)*. The void fraction follows the same trend, with a rapid decrease as the flow transitions from two-phase to liquid-dominated conditions.

The comparison between the two geometries shows that the condensation behavior is nearly identical. Both designs reach saturated liquid conditions at the same location (segment 6), and the difference in vapor quality evolution is minimal. The modified design exhibits slightly lower quality values in the upstream segments, indicating a marginally faster condensation rate; however, this difference remains very small and does not affect the overall condensation length. This confirms that the change in tube geometry has a negligible effect on the condensation process.

This observation is further supported by the overall condenser results summarized in *Table 3*. Both designs produce essentially identical outlet conditions, with a vapor quality of approximately 1×10^{-5}, a void fraction of about 0.00219, and a total heat transfer rate of 339.66 W. These results indicate that the energy removal from the refrigerant is unchanged, suggesting that the overall heat transfer is not strongly influenced by the refrigerant-side geometry under the given operating conditions. Instead, the air-side thermal resistance likely dominates the heat transfer process, limiting the impact of changes in passage diameter and number.

In contrast, the pressure results show a clear difference between the two designs. The pressure decreases along the condenser in both cases due to two-phase frictional effects *(Fig. 5)*, but the modified design consistently exhibits a slightly lower pressure at each segment. The total pressure drop increases from 0.12283 kPa in the original design to 0.26170 kPa in the modified design. This behavior is physically consistent, as the smaller passage diameter reduces the total flow area, increases the mass flux, and leads to higher frictional losses.

Although the pressure drop approximately doubles in relative terms, the absolute increase remains small (approximately 0.14 kPa). Therefore, while the hydraulic penalty is clearly captured by the model, its practical significance is limited.

Overall, the results indicate that the proposed modification provides no meaningful improvement in condensation performance while introducing a small increase in pressure drop. Since the thermal performance remains unchanged and the pressure loss is slightly higher, the modified design does not offer a clear advantage. Under the specified operating conditions, the original condenser design (9 passages, 3.0 mm diameter) is therefore preferable.

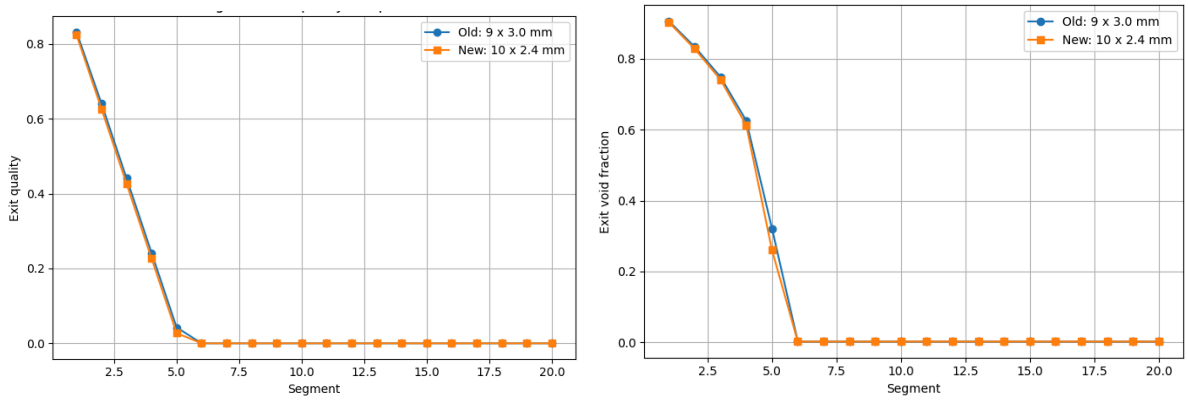

Figure 4 | *Comparison between old and new configurations of segment exit quality (left) and segment exit void fraction.*

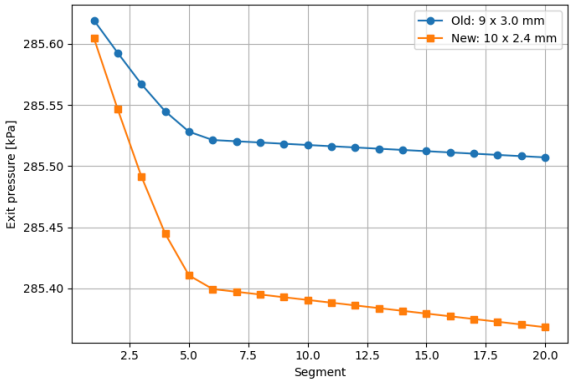

Figure 5 | *Comparison between old and new configurations of segment exit pressure.*

Table 3 | *Comparison of condenser outlet conditions and performance for the original and modified geometries.*

Quantity	Old (9 x 3.0 mm)	New (10 x 2.4 mm)
Outlet pressure [kPa]	285.50717	285.36830
Outlet quality [-]	0.000010	0.000010
Outlet void fraction [-]	0.002186	0.002186
Total heat transfer to air [W]	339.657	339.657
Total pressure drop [kPa]	0.12283	0.26170
Segment where x ≈ 0	6	6

4.2 Tube Length Adjustment for Complete Condensation

To ensure that the condenser produces exactly saturated liquid at its outlet, the required condenser tube length was determined for the modified geometry (10 passages, 2.4 mm diameter). In the previous analysis, the condenser length was fixed at 20 segments (total length 0.80 m), which resulted in complete condensation occurring before the final segment. This indicates that the condenser is longer than necessary for the given operating conditions.

4.2.1 Methodology

To address this, the model was adapted to identify the precise location along the condenser where the vapor quality reaches zero. Rather than restricting the analysis to integer numbers of segments, the calculation was extended to allow for partial segment lengths. When the vapor quality was predicted to cross zero within a segment, linear interpolation was applied to determine the exact fraction of that segment required to achieve x=0. This approach provides a more accurate estimate of the minimum condenser length needed for complete condensation.

The complete Python code can be found in *Appendix C2*.

4.2.2 Results & discussion

The results show that the required condenser tube length is:

- 0.20550 m (20.550 cm)
- Equivalent to approximately 5.138 segments of length 0.040 m

The segment-by-segment results indicate that condensation progresses steadily through the condenser, with the vapor quality decreasing from 0.999 at the inlet to approximately 0.0272 at the end of segment 5. Complete condensation occurs within segment 6, where only a small fraction of the segment (dz≈0.00550 m) is required to reach saturated liquid conditions.

These results demonstrate that the originally assumed condenser length (20 segments, 0.80 m) is significantly longer than necessary. In fact, only about 26% of the original length is required to fully condense the refrigerant under the specified operating conditions. This indicates that the condenser is substantially oversized.

From a design perspective, reducing the condenser length to the minimum required value would offer several advantages, including reduced material usage, lower weight, and potentially lower cost, without compromising thermal performance. However, it should also be noted that maintaining some excess length may be desirable in practice to provide a safety margin for variations in operating conditions, such as increased heat load or higher inlet air temperature.

Overall, this analysis shows that the modified condenser geometry achieves complete condensation well before the end of the originally specified length, and that the optimal tube length for the given conditions is approximately 0.21 m.

4.3 System Adjustment for Increased Air Inlet Temperature

To evaluate the robustness of the modified condenser design, the inlet air temperature was increased from 20 °C to 25 °C while maintaining the geometry determined in Task 3(a) and the optimized tube length from Task 3(b) (0.2056 m). The objective was to identify one or more system modifications that would still allow the system to meet both the required chip heat dissipation rate of 340 W and the target chip temperature of less than 55 °C.

4.3.1 Methodology

The existing condenser model was extended to simulate performance under the new inlet air temperature. The baseline Task 3 system was first evaluated using:

- 10 refrigerant passages of 2.4 mm diameter
- tube length of 0.20550 m (from Task 3(b))
- air velocity of 6.5 m/s
- evaporator pressure of 285.63 kPa
- inlet air temperature of 25 °C

The condenser was simulated segment-by-segment, allowing for a partial final segment to accurately capture the point at which condensation would be completed. The model computed the outlet quality, outlet pressure, total heat transfer to the air, and predicted chip temperature based on the evaporator pressure.

Since the baseline configuration did not achieve full condensation at the higher air temperature, a parametric search was conducted to identify feasible system modifications. The following parameters were varied:

- evaporator pressure (to adjust saturation temperature and heat transfer driving force)
- air velocity (to modify air-side heat transfer coefficient)
- condenser tube length (to increase heat transfer area and residence time)

For each combination, the system was evaluated to determine whether it satisfied the design criteria:

- outlet quality $x \approx 0$ (fully condensed liquid)
- total heat transfer ≈ 340 W
- chip temperature < 55 °C

A scoring metric was used to identify solutions that required minimal deviation from the baseline design.

The complete Python code can be found in *Appendix C3*.

4.3.2 Results & discussion

The baseline Task 3 system at 25 °C was found to be insufficient to fully condense the refrigerant. The predicted outlet quality was 0.1549, indicating that a significant fraction of vapor remains at the condenser outlet. Correspondingly, the total heat transfer is 286.99 W, which is below the required 340 W. In addition, the predicted chip temperature is 56.20 °C, exceeding the target limit of 55 °C.

This reduction in performance is expected due to the decreased temperature difference between the refrigerant saturation temperature and the inlet air at higher air temperature, which reduces the heat transfer driving force. As a result, the condenser becomes the limiting component, preventing both complete condensation and sufficient heat removal.

Several feasible system modifications were identified that restore full condensation and meet all design constraints. One representative solution is summarized in *Table 4*. In this configuration:

- the evaporator pressure is reduced from 285.63 kPa to 270 kPa
- the air velocity is increased from 6.5 m/s to 7.0 m/s

- the condenser tube length is increased from 0.2055 m to 0.3282 m

This modified system achieves complete condensation at the outlet (x=0), restores the total heat transfer to approximately 339.66 W, and reduces the chip temperature to 53.99 °C, which satisfies the design requirement.

Table 4 | *Comparison of baseline and modified system at Ta,in=25°C.*

Quantity	Unaltered system @ 25°C	Altered feasible system @ 25°C
Evaporator pressure [kPa]	285.63	270.00
Air velocity [m/s]	6.5	7
Tube length [m]	0.20550	0.32823
Tube length [cm]	20.550	32.823
Inlet air temperature [°C]	25	25
Outlet pressure [kPa]	285.37922	269.75580
Outlet quality [-]	0.15490	0
Outlet void fraction [-]	0.53969	0
Total heat transfer to air [W]	286.99	339.66
Predicted chip temperature [°C]	56.20	53.99

The improvement in performance can be explained by a combination of effects. Increasing the air velocity enhances the air-side heat transfer coefficient, which is the dominant thermal resistance in the condenser. This directly increases the overall heat transfer rate. Additionally, increasing the condenser length provides more surface area and residence time, allowing the refrigerant to fully condense.

Interestingly, a slight reduction in evaporator pressure is also observed in the optimal solution. While this reduces the refrigerant saturation temperature, the combined effect of increased air-side convection and extended condenser length compensates for this, enabling the system to meet the required heat dissipation rate.

The required modifications are modest. The tube length increases by approximately 3 cm, and the air velocity increases slightly, both of which are practical adjustments. The reduction in pressure is also moderate and does not introduce significant design complications.

Overall, the analysis shows that the modified condenser design can meet the required performance at elevated air inlet temperatures, provided that small adjustments to operating conditions, particularly air velocity and condenser length, are implemented.

5. Task 4 – Nucleate Boiling in the Evaporator and Critical Heat Flux

5.1 Problem Description

The critical heat flux (CHF) acts as the performance limit for nucleate boiling heat transfer. After which, the heat transfer coefficient decreases because the vapour can form a film that can cover the contact area with the hot object and insulate it in a layer of vapour which can trap heat and cause the chip to overheat (film boiling region). Therefore, it is important that we determine the CHF or the maximum heat flux of our system when designing an electronics cooling system. In Task 4, we are asked to calculate the maximum heat flux possible at each of the following evaporator pressures: P2 = 160, 180, 200, 220 and 240 kPa. In Part (a), we are asked to determine the correct method to predict the maximum heat flux based on the bubble length scale that applies to each case. Then, calculate the maximum heat flux for each case. This helps understand if the target design heat flux exceeds the predicted critical heat flux at any of the above pressures. A plot is to be provided of the critical heat flux versus the pressure for this system.

In Part (b), we are asked to determine the required chip temperature to achieve the maximum heat flux (higher than 340 W) associated with the design pressure of 285.63 kPa found in Task 2. As well, in the case that the chip is designed to operate at this temperature for a limited period of time, we are asked to propose changes to the condenser design to accommodate this. For our proposed operating conditions, we are asked to provide a table detailing how the quality, heat transfer rate, and pressure vary along the tube. As well, to report the condenser's total refrigerant heat transfer and pressure drop values.

In Part (c), the heat output of the chip varies from 300 W to 650 W during operation. For the design we recommended in Part (b), we are asked to plot the variation of the minimum air velocity with chip heat output that would keep the chip temperature below 65 degrees C. In the case that this temperature cannot be achieved for the conditions in Part (b), we are asked to report the highest chip heat dissipation rate that can achieve a chip temperature of 65 degrees C and the minimum air velocity to get to a chip temperature of 65 degrees C at that heat output rate.

5.2 Thermophysical Properties

For nucleate boiling, Zuber (1959) recommends the following relation to find the maximum critical heat flux $q''_{max,Z}$.

$$q''_{max,Z} = 0.131\rho_v h_{lv}\left[\frac{\sigma(\rho_l-\rho_v)g}{\rho_v^{\,2}}\right]^{1/4}$$

$$L_b = \sqrt{\frac{\sigma}{g(\rho_l-\rho_v)}}$$

Note that L_b is a bubble length scale that indicates the departure size of the bubble during nucleate boiling. The most dangerous wavelength associated with Taylor instability which can dictate vapour column spacing during boiling on upward-facing surfaces is as follows:

$$\lambda_D = (2\pi\sqrt{3})L_b$$

When nucleate boiling on an upward facing surface approaches the critical heat flux condition, the space between the vapor columns scales with λ_D and as a result with L_b. Lienhart and Dhir (1970-73) argued that the critical heat flux deviates from the Zuber prediction, when the size of the surface becomes comparable to L_b. So, per the following, we can predict the critical or maximum heat flux (q''_{max}).

$$q''_{max} = 1.14q''_{max,Z} \text{ for } \frac{L}{L_b} > 30$$

$$q''_{max} = \frac{1.14\lambda_D^{\ 2}}{A_{heater}} q''_{max,Z} \text{ for } 9 < \frac{L}{L_b} < 20$$

5.3 Methodology and Results

5.3.1 Task 4 Part (a) - Calculating the Critical Heat Flux (CHF)

In Part (a), we determined the correct method to predict the maximum heat flux based on the bubble length scale that applies to each case by calculating the bubble length scale L_b in the program loop for each pressure value. Then, employing an if statement in the program that selected the correct method for critical heat flux determination, we calculated the maximum heat flux for each case (see Appendix D). The calculated values were plotted and can be seen in Figure 6 below.

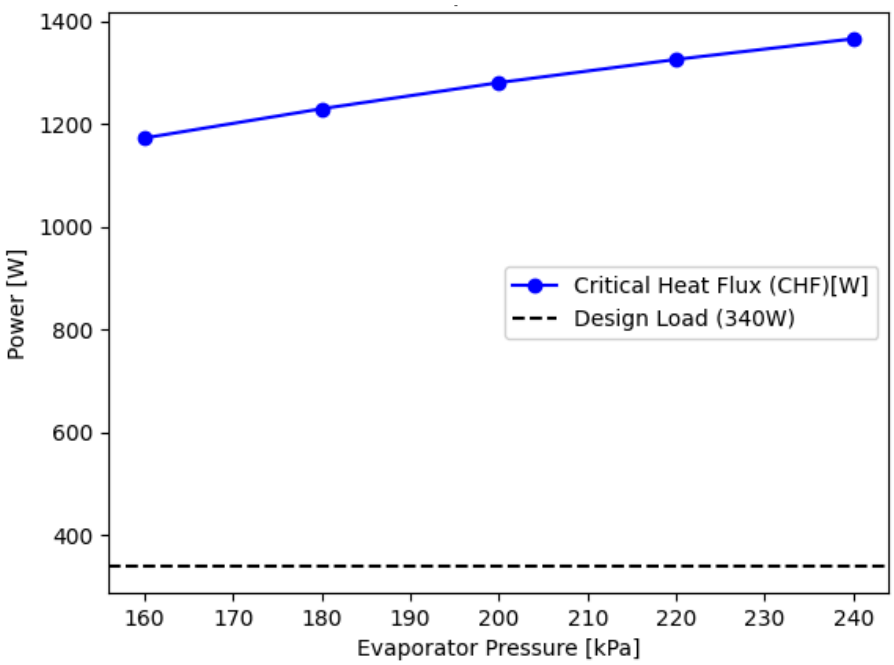

Figure 6 | *Critical Heat Flux (CHF) versus the Pressure Value, compared to the Design Chip Heat Flux of 340 W.*

5.3.2 Task 4 Part (b) - Exploring the Impact of CHF on Temperature and Designing a Better Condenser based on New CHF Limits

As a reminder, assuming negligible thermal resistance at the evaporator wall allows us to assume $T_{chip} = T_w$. Then, we use the "T_wall_calculator" function that was defined in the program for Task 2 (see Appendix D for full Task 4 code) to find the chip temperature at the CHF. The CHF corresponding to the design pressure of 285.63 kPa found in Task 2, is found as 1441.315 W. Then, we have all the required inputs to utilize the "T_wall_calculator" function and we find that the required chip temperature to achieve the critical heat flux of 1441.315 W (higher than 340 W) associated with the design pressure of 285.63 kPa is 53.03 °C.

We analyzed changes to the condenser design to accommodate the chip operating at this temperature for a limited period of time. The investigated cases can be seen in Table 5 below.

Table 5 | *Case Description*

Case Number	Description
1	Increasing the condenser air flow velocity by 90%.
2	Increasing the evaporator pressure (P2) by 30% more than the design pressure and altering the number of the circular flow passages in the flat tube to 12.
3	Increasing the condensing tube length (number of sections to 30).
4	Increasing the condensing tube length (number of sections to 24). And increasing the evaporator pressure (P2) by 30% more than the design pressure. And altering the number of the circular flow passages in the flat tube to 14.

Table 6 and *Figure 7* below detail how the quality, heat transfer rate, and pressure vary along the tube for different condenser designs and report the cumulative heat transfer and pressure drop values for the refrigerant.

Table 6 | *Summary of Cumulative Properties for the Full Condenser Length*

Cumulative Properties	**Value**	**Unit**	**Cumulative Properties**	**Value**	**Unit**	**Cumulative Properties**	**Value**	**Unit**	**Cumulative Properties**	**Value**	**Unit**
Case 1			**Case 2**			**Case 3**			**Case 4**		
Total Refrigerant Pressure Drop (ΔP)	1.0240	kPa	Total Refrigerant Pressure Drop (ΔP)	0.7651	kPa	Total Refrigerant Pressure Drop (ΔP)	1.9621	kPa	Total Refrigerant Pressure Drop (ΔP)	0.6933	kPa
Total Rate of Heat Transfer of Condenser (q_dot)	1439.86	W	Total Rate of Heat Transfer of Condenser (q_dot)	1439.86	W	Total Rate of Heat Transfer of Condenser (q_dot)	1439.86	W	Total Rate of Heat Transfer of Condenser (q_dot)	1439.86	W
Exit State of R-245fa out of the Condenser: Saturated Liquid (Fully Condensed)			Exit State of R-245fa out of the Condenser: Saturated Liquid (Fully Condensed)			Exit State of R-245fa out of the Condenser: Saturated Liquid (Fully Condensed)			Exit State of R-245fa out of the Condenser: Saturated Liquid (Fully Condensed)		
Exit: Outlet pressure P_{out}	284.6076	kPa	Exit: Outlet pressure P_{out}	370.5561	kPa	Exit: Outlet pressure P_{out}	283.6696	kPa	Exit: Outlet pressure P_{out}	370.6278	kPa
Exit: Outlet vapor quality x_{out}	0.000010	-	Exit: Outlet vapor quality x_{out}	0.000010	-	Exit: Outlet vapor quality x_{out}	0.000010	-	Exit: Outlet vapor quality x_{out}	0.000010	-

[Part b] Case 1 Condenser Simulation Results

Seg	x_out	α_out	P_out (kPa)	q_seg (W)
1	0.92802	0.94689	285.5423	102.3061
2	0.85592	0.91568	285.4270	103.9227
3	0.78375	0.88785	285.3162	104.0131
4	0.71160	0.86093	285.2127	103.9969
5	0.63947	0.83368	285.1174	103.9643
6	0.56736	0.80515	285.0306	103.9270
7	0.49528	0.77441	284.9524	103.8887
8	0.42323	0.74028	284.8824	103.8502
9	0.35120	0.70116	284.8204	103.8117
10	0.27920	0.65447	284.7663	103.7721
11	0.20724	0.59559	284.7199	103.7295
12	0.13530	0.51468	284.6814	103.6792
13	0.06338	0.38280	284.6517	103.6604
14	0.00001	0.00219	284.6334	91.3376
15	0.00001	0.00219	284.6291	0.0000
16	0.00001	0.00219	284.6248	0.0000
17	0.00001	0.00219	284.6205	0.0000
18	0.00001	0.00219	284.6162	0.0000
19	0.00001	0.00219	284.6119	0.0000
20	0.00001	0.00219	284.6076	0.0000

[Part b] Case 2 Condenser Simulation Results

Seg	x_out	α_out	P_out (kPa)	q_seg (W)
1	0.93654	0.94654	371.2654	90.0255
2	0.87338	0.91583	371.1921	91.0378
3	0.81013	0.88892	371.1199	91.1640
4	0.74687	0.86339	371.0506	91.1736
5	0.68361	0.83813	370.9851	91.1683
6	0.62037	0.81240	370.9239	91.1582
7	0.55713	0.78555	370.8671	91.1459
8	0.49390	0.75692	370.8147	91.1326
9	0.43068	0.72570	370.7669	91.1186
10	0.36747	0.69083	370.7237	91.1040
11	0.30428	0.65074	370.6850	91.0886
12	0.24109	0.60294	370.6509	91.0717
13	0.17792	0.54292	370.6216	91.0520
14	0.11476	0.46112	370.5973	91.0268
15	0.05163	0.32936	370.5789	90.9881
16	0.00001	0.00199	370.5679	74.4037
17	0.00001	0.00199	370.5649	0.0000
18	0.00001	0.00199	370.5620	0.0000
19	0.00001	0.00199	370.5591	0.0000
20	0.00001	0.00199	370.5561	0.0000

[Part b] Case 3 Condenser Simulation Results

Seg	x_out	α_out	P_out (kPa)	q_seg (W)
1	0.95225	0.95930	285.5139	67.3759
2	0.90516	0.93635	285.3689	67.8746
3	0.85797	0.91652	285.2246	68.0249
4	0.81078	0.89811	285.0842	68.0077
5	0.76362	0.88036	284.9493	67.9769
6	0.71648	0.86281	284.8205	67.9427
7	0.66936	0.84516	284.6984	67.9079
8	0.62227	0.82712	284.5830	67.8735
9	0.57520	0.80846	284.4746	67.8401
10	0.52816	0.78891	284.3730	67.8079
11	0.48113	0.76817	284.2784	67.7771
12	0.43413	0.74590	284.1906	67.7477
13	0.38714	0.72164	284.1096	67.7196
14	0.34018	0.69481	284.0353	67.6928
15	0.29323	0.66458	283.9678	67.6670
16	0.24630	0.62973	283.9069	67.6420
17	0.19939	0.58838	283.8529	67.6173
18	0.15249	0.53726	283.8060	67.5921
19	0.10560	0.46997	283.7664	67.5850
20	0.05873	0.37089	283.7351	67.5544
21	0.01189	0.17057	283.7140	67.5069
22	0.00001	0.00219	283.7040	17.1254
23	0.00001	0.00219	283.6997	0.0000
24	0.00001	0.00219	283.6954	0.0000
25	0.00001	0.00219	283.6911	0.0000
26	0.00001	0.00219	283.6868	0.0000
27	0.00001	0.00219	283.6825	0.0000
28	0.00001	0.00219	283.6782	0.0000
29	0.00001	0.00219	283.6739	0.0000
30	0.00001	0.00219	283.6696	0.0000

Figure 7 *| Summary of Exploratory Simulation Results for the Full Condenser Length - Quality (x), Void Fraction (α), Pressure at the Outlet of each Segment, and the Rate of Heat Transfer. Case 1, Case 2, and Case 3.*

Based on the data in Table 6, the best performing case is Case 4 as it has the lowest total pressure drop and the same total heat transfer rate as the other alternatives. Its operating conditions are as follows.

Operating conditions of Case 4: **(changes to the conditions are bolded)**

- CHF Heat load: $\dot{q}_{CHF}$ = **1441.315 W**
- Chip Temperature corresponding to the new CHF Heat Load= **53.025 C**
- Air inlet temperature: $T_{a,in}$ = 20°C
- Air velocity: V_a = 6.5 m/s
- Length of each segment: Δz = 0.04 m
- Number of segments: **i = 24**
- Number of parallel passages: **n = 14**
- Evaporator Pressure (Input Pressure to the Condenser) $= 285.63 \times 1.3 =$ **371.319 kPa**
- Tube (passage) diameter: D = 3 mm = 0.003 m
- $\theta = 35\ °C$
- Area of chip: A_{chip} = 0.0036 m
- Inlet vapor quality: x_{s0} = 0.999
- Inlet void fraction: $\alpha_{s0} = 0.999$

For this proposed best new condenser design, Figure 8 below shows the details of how the phase, hydraulic, and thermal properties change along the condenser tube for Case 4, the proposed best new condenser design.

Seg	x_out	α_out	P_out (kPa)	q_seg (W)
1	0.93667	0.94661	371.2752	89.8319
2	0.87359	0.91593	371.2122	90.9306
3	0.81034	0.88900	371.1491	91.1568
4	0.74708	0.86347	371.0881	91.1746
5	0.68382	0.83821	371.0300	91.1733
6	0.62057	0.81248	370.9752	91.1658
7	0.55733	0.78563	370.9240	91.1557
8	0.49409	0.75700	370.8765	91.1440
9	0.43086	0.72578	370.8328	91.1315
10	0.36764	0.69092	370.7931	91.1180
11	0.30444	0.65084	370.7573	91.1035
12	0.24124	0.60305	370.7257	91.0873
13	0.17805	0.54306	370.6984	91.0681
14	0.11489	0.46130	370.6757	91.0432
15	0.05175	0.32968	370.6584	91.0045
16	0.00001	0.00199	370.6479	74.5707
17	0.00001	0.00199	370.6454	0.0000
18	0.00001	0.00199	370.6429	0.0000
19	0.00001	0.00199	370.6404	0.0000
20	0.00001	0.00199	370.6379	0.0000
21	0.00001	0.00199	370.6354	0.0000
22	0.00001	0.00199	370.6329	0.0000
23	0.00001	0.00199	370.6303	0.0000
24	0.00001	0.00199	370.6278	0.0000

Figure 8 | *Summary of the Proposed Case 4 Results for the Full Condenser Length - Quality (x), Void Fraction (α), Pressure at the Outlet of each Segment, and the Rate of Heat Transfer.*

5.3.3 Task 4 Part (c) - Varying Chip Heat Output and Corresponding Air Velocity

For a chip with varying heat output from 300 W to 650 W and the Case 4 cooling design we recommended in Part (b), we plotted the variation of the minimum air velocity with chip heat output that would keep the chip temperature below 65 °C. See *Figure 9* below.

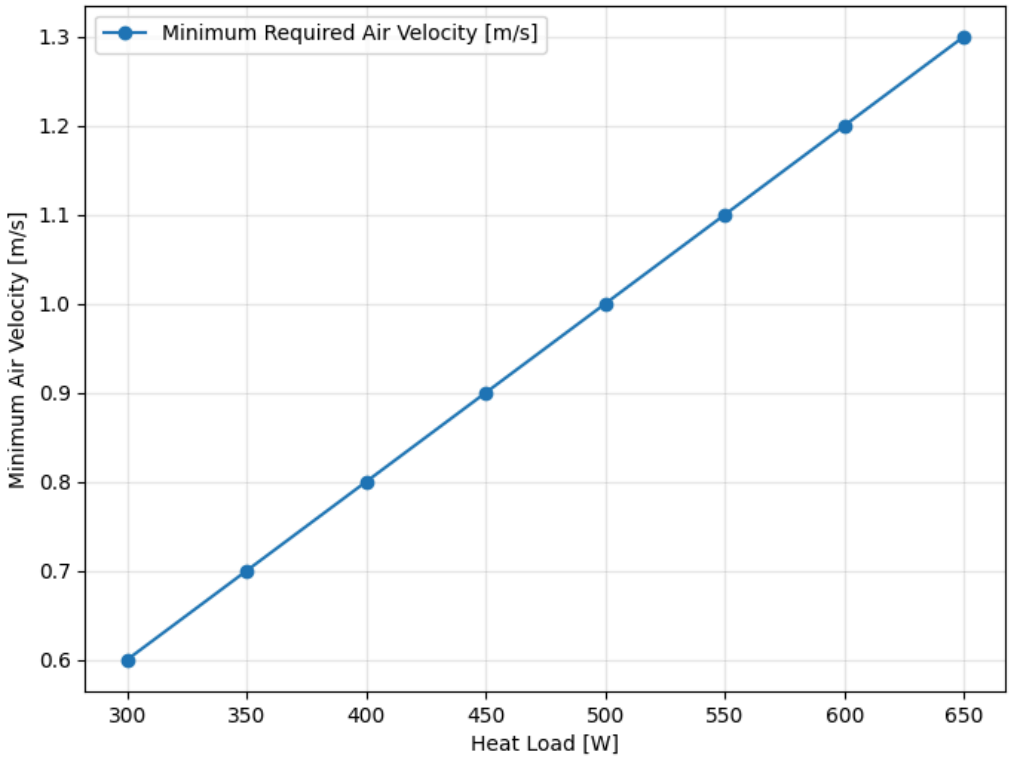

Figure 9 *| Minimum Air Velocity [m/s] Required to Maintain* $T_{chip} < 65\,°C$ *versus Varying Chip Heat Output [W].*

To find the minimum air velocity required to maintain the temperature of the chip below 65 °C, we employed a for loop. This for loop, iterated through the heat load values 300-650 W, and first calculated what the chip temperature would be at various different pressure values (100-500 kPa) for each heat load value. This allowed us to map a curve for the relationship between the evaporator pressure and chip temperature values for the new condenser design (Case 4 from Part b). Then, we interpolated to find the value of the evaporator pressure (the pressure limit) at which the temperature of 65 °C can be achieved for the chip. This pressure limit was found as 377.96 kPa (see *Figure 10* below).

Next, we defined a test starting value for the air velocity to be used in the exploration loop, and ran the condenser simulation to ensure that at the pressure limit found above and with the test air velocity value, the condenser was able to produce saturated liquid at its exit. This was verified by checking: condenser exit quality = 0. Using a while loop, the test air velocity value was incrementally increased (+0.2 m/s) for each q value until it reached a value of 20 m/s (a very high value that would not be reasonable for this application) to allow for a large exploration range. If the test value reached that limit, the loop would break and we would be informed that the design is not solvable for any air velocity. This did not occur at any of the heat load values. So, the minimum required air velocity values for each heat load value that was simulated for a constant operating chip temperature of 65 C are reported below in Table 7. Note that the code can be seen in *Appendix D*.

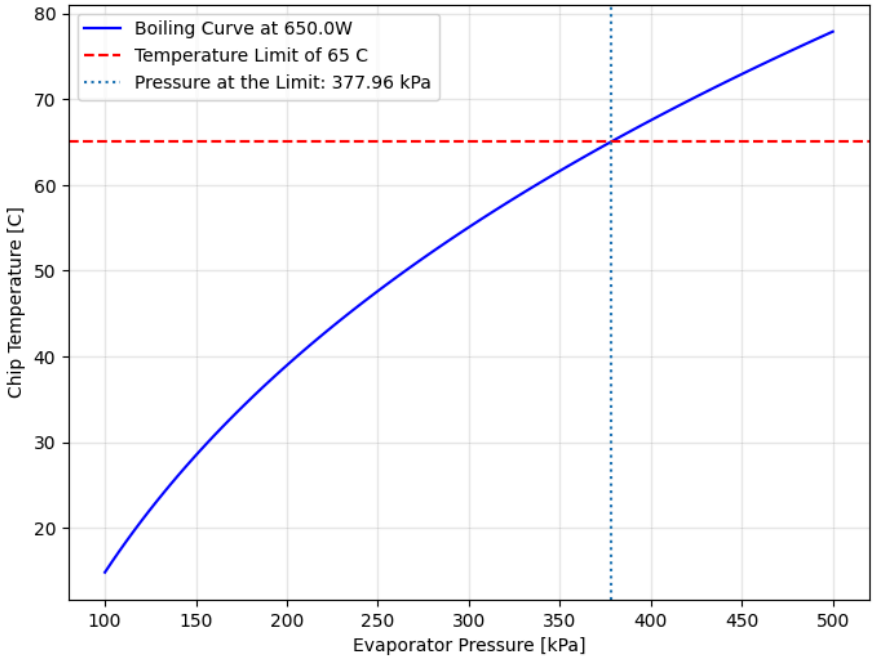

Figure 10 *| Interpolation at 650W Heat Load to Find the Pressure Limit at* $T_{chip} = 65\ °C$.

Table 7 *| Minimum Air Velocity Required for T_chip = 65 °C at Various Chip Heat Loads with Corresponding Pressure Limit.*

Heat Load [W]	Pressure Limit [kPa]	Minimum Air Velocity [m/s]
300	377.96	0.6
350	377.96	0.7
400	377.96	0.8
450	377.96	0.9
500	377.96	1.0
550	377.96	1.1
600	377.96	1.2
650	377.96	1.3

5.4 Discussion

Based on the results found in Part (a), in *Figure 6*, the target design heat flux of 340 W does not exceed the predicted CHF at any of these pressures. This tells us that heat loads around 340 W are safe for this system since there is a large safety margin between the heat load and the CHF for the system. Assuring us that the evaporator can remain in the nucleate boiling region while transferring heat and not enter the film boiling region where the temperature of the chip can rise due to the insulating effect of the vapour film covering the contact surface between the chip and the liquid refrigerant.

In Part (b), we found that the required chip temperature to achieve the critical heat flux of 1441.315 W (higher than 340 W) associated with the design pressure of 285.63 kPa is 53.03 °C. Next, we compared the performance of various different design changes to the condenser and found that Case 4's combination of operating conditions performed best if we were to operate the chip and the cooling system at a temperature of 53.03 °C for a longer period of time. We noticed that by increasing the number of segments of the condenser (lengthening it) and by increasing the number of parallel passages (tubes in the cross section) while increasing the inlet pressure of the condenser, we can achieve a lower pressure drop compared to other alternatives where the pressure at the inlet was not increased and was held constant at the old evaporator design pressure. This comparison of condenser design cases is conducted in Table 6 above. Indicating that Case 4 has the lowest total pressure drop for the system's newly altered CHF of 1441.315 W and that Case 4 maintains the same total heat transfer rate as the other alternatives at 1439.86 W due to minor losses and varying air temperature and humidity that is not captured by the model. All designs were able to achieve a saturated liquid state at the end of the condenser, making them viable options.

In Part (c), the heat output of the chip varies from 300 W to 650 W during operation. For the design we recommended in Part (b), we found and plotted *(Figure 9)* the variation of the minimum air velocity with chip heat output that would keep the chip temperature below 65 °C. It can be seen in Figure 9 that the minimum air velocity required to maintain the chip operating temperature below 65 °C is increasing almost linearly with the increase in the heat load on the cooling system. This makes sense since with the increasing heat load on the cooling system, there is more heat energy that must be dissipated in the evaporator. So, in order to keep the chip temperature around and below 65 °C, the saturation pressure in the evaporator must decrease, to allow for easier and more rapid evaporation. This means we would need a larger mass flow rate to dissipate the heat energy because the refrigerant density is lowered. In this cooling system, we know that the ambient air temperature cannot be changed and condenser design was selected in Part b. Therefore, the only possible avenue for the cooling system to dissipate the increased heat energy is to ramp up the flow rate or the velocity of the air in the same manner that the heat energy increases (linearly).

6. Distribution of work

Task 1	Marie-Batisse
Task 2	Asal
Task 3	Marie-Batisse
Task 4	Asal
Write-up	Each for corresponding tasks

Appendices

Appendix A1 - Lagrange interpolation for Thermophysical Properties

To evaluate the thermophysical properties of R245fa at the operating pressure, saturation temperature was first determined from the pressure–temperature relation provided in the project statement. Since the required property values do not generally coincide exactly with the tabulated temperatures, Lagrange interpolation was used to estimate the saturated liquid and vapor properties at the calculated saturation temperature.

This interpolation procedure was adapted from Project 1 and applied here to the R245fa property table provided for Project 2.

The interpolation uses the tabulated saturation data for R245fa:

- Temperature, T [°C]
- Saturated liquid density, ρl [kg/m3]
- Saturated vapor density, ρv [kg/m3]
- Saturated liquid viscosity, μl [Pa·s]
- Saturated vapor viscosity, μv [Pa·s]
- Saturated liquid thermal conductivity, kl [W/m·K]
- Latent heat of vaporization, hlv [J/kg]
- Surface tension, σ [N/m]
- Saturated liquid specific heat, cp,l [J/kg·K]

For the inlet pressure of Task 1, $P_{in} = 220\ kPa$, the saturation temperature is obtained from:

$$T_{sat} = \frac{C_1}{ln(P) - C_0}$$

with C_0 = 20.472 and C_1 = −2579.9, where pressure is in Pa.

Python code

```
#Appendix A1 - Lagrange interpolation for Thermophysical Properties

import numpy as np

# ----------------------------
# R245fa saturation property data
# ----------------------------
T_data_C = np.array([15.1, 30.0, 46.0, 60.0])          # °C

rho_l_data = np.array([1365.0, 1325.0, 1279.0, 1267.0])      # kg/m^3
rho_v_data = np.array([5.961, 10.169, 17.30, 25.68])         # kg/m^3
mu_l_data  = np.array([464e-6, 382e-6, 314e-6, 266e-6])      # Pa·s
mu_v_data  = np.array([9.93e-6, 10.4e-6, 11.0e-6, 11.5e-6])  # Pa·s
k_l_data   = np.array([0.0912, 0.0865, 0.0814, 0.0769])      # W/m·K
h_lv_data  = np.array([196.1e3, 187.0e3, 177.2e3, 168.0e3])  # J/kg
```

```python
sigma_data = np.array([0.0153, 0.0134, 0.0114, 0.0096])       # N/m
cp_l_data  = np.array([1318.0, 1350.0, 1388.0, 1428.0])       # J/kg·K

# Tsat(P) constants
C0 = 20.472
C1 = -2579.9


def lagrange_interp(x, xdata, ydata):
    n = len(xdata)
    y = 0.0
    for i in range(n):
        Li = 1.0
        for j in range(n):
            if i != j:
                Li *= (x - xdata[j]) / (xdata[i] - xdata[j])
        y += ydata[i] * Li
    return y


def Tsat_from_P(P_kPa):
    P_Pa = P_kPa * 1000.0
    Tsat_K = C1 / (np.log(P_Pa) - C0)
    return Tsat_K


def r245fa_sat_props(P_kPa):
    Tsat_K = Tsat_from_P(P_kPa)
    Tsat_C = Tsat_K - 273.15

    rho_l = lagrange_interp(Tsat_C, T_data_C, rho_l_data)
    rho_v = lagrange_interp(Tsat_C, T_data_C, rho_v_data)
    mu_l  = lagrange_interp(Tsat_C, T_data_C, mu_l_data)
    mu_v  = lagrange_interp(Tsat_C, T_data_C, mu_v_data)
    k_l   = lagrange_interp(Tsat_C, T_data_C, k_l_data)
    h_lv  = lagrange_interp(Tsat_C, T_data_C, h_lv_data)
    sigma = lagrange_interp(Tsat_C, T_data_C, sigma_data)
    cp_l  = lagrange_interp(Tsat_C, T_data_C, cp_l_data)

    Pr_l = mu_l * cp_l / k_l

    return {
        "Tsat_K": Tsat_K,
        "Tsat_C": Tsat_C,
        "rho_l": rho_l,
        "rho_v": rho_v,
        "mu_l": mu_l,
        "mu_v": mu_v,
        "k_l": k_l,
        "h_lv": h_lv,
        "sigma": sigma,
        "cp_l": cp_l,
        "Pr_l": Pr_l
    }


# Example evaluation for Task 1 inlet pressure
props = r245fa_sat_props(220.0)

units = {
    "Tsat_K": "K",
    "Tsat_C": "°C",
    "rho_l": "kg/m^3",
    "rho_v": "kg/m^3",
    "mu_l": "Pa·s",
    "mu_v": "Pa·s",
    "k_l": "W/m·K",
    "h_lv": "J/kg",
    "sigma": "N/m",
    "cp_l": "J/kg·K",
    "Pr_l": "-"
}
```

```
for key, val in props.items():
   print(f"{key:8s} = {val:.6f} {units[key]}")
```

Output

```
Tsat_K   = 315.753381 K
Tsat_C   = 42.603381 °C
rho_l    = 1287.103727 kg/m^3
rho_v    = 15.564896 kg/m^3
mu_l     = 0.000327 Pa·s
mu_v     = 0.000011 Pa·s
k_l      = 0.082487 W/m·K
h_lv     = 179320.396169 J/kg
sigma    = 0.011826 N/m
cp_l     = 1379.389562 J/kg·K
Pr_l     = 5.469024 -
```

Appendix A2 - Single Segment Condenser Element

The function evaluates a single condenser segment by combining heat transfer and pressure drop models. The outlet conditions of this segment serve as inputs for subsequent segments in Task 2.

Inputs

Geometry:

- D=3 mm
- Δz=0.04 m
- n=9

Operating conditions:

- Pin=220 kPa
- xin=0.999
- $\dot{q}_{chip}$=340 W

Air side:

- Ta,in=20°C
- Va=6.5 m/s
- ρa=1.3 kg/m3
- cp=1005 J/kgK

Python code

```
#Appendix A2 - Single Segment Condenser Element

def segment_model(
   q_chip=340.0,        # W
   Ta_in_C=20.0,        # °C
```

```
    Va=6.5,              # m/s
    P_in_kPa=220.0,      # kPa
    dz=0.040,            # m
    D=0.003,             # m
    n_tubes=9,           # -
    x_in=0.999,          # -
    rho_a=1.3,           # kg/m^3
    cp_a=1005.0,         # J/kg·K
    Lf=5.26e-3,          # m
    g=9.81               # m/s^2
):
    props = r245fa_sat_props(P_in_kPa)

    Tsat_K = props["Tsat_K"]
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]
    k_l = props["k_l"]
    h_lv = props["h_lv"]
    sigma = props["sigma"]
    Pr_l = props["Pr_l"]

    Ta_in_K = Ta_in_C + 273.15

    # Mass flux
    A_passage = np.pi * D**2 / 4.0                        # m^2
    A_total = n_tubes * A_passage                         # m^2
    G = q_chip / (A_total * h_lv)                         # kg/m^2·s

    # Reynolds numbers
    Re_lo = G * D / mu_l
    Re_l = G * (1.0 - x_in) * D / mu_l
    Re_vo = G * D / mu_v

    # Martinelli parameter
    X_tt = ((1.0 - x_in) / x_in)**0.9 * (rho_v / rho_l)**0.5 * (mu_l / mu_v)**0.1

    # Traviss F_T
    if Re_l > 1125:
        F_T = 0.812 * (Pr_l**0.5) + 5.0 * np.log(1.0 + 5.0 * Pr_l) + 2.5 * np.log(0.0031 *
Re_l)
    elif Re_l > 50:
        F_T = 0.585 * (Pr_l**0.5) + 5.0 * np.log(1.0 + Pr_l * (0.0964 * Re_l**0.585 - 1.0))
    else:
        F_T = 0.5 * (Pr_l**0.5) * Re_l

    # Refrigerant-side heat transfer coefficient
    hr = (k_l / D) * 0.15 * (Pr_l**0.4) * (Re_l**0.9) * F_T * (1.0 / X_tt + 2.85 /
X_tt**0.476)

    # Inlet void fraction
    alpha_in = 1.0 / (
        1.0
        + ((1.0 - x_in) / x_in)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Segment UA and heat transfer
    Ar = n_tubes * np.pi * D * dz                         # m^2
    UA_air = 77.1 * (Va / 6.5)**0.5 * dz                 # W/K
    UA_seg = 1.0 / (1.0 / (hr * Ar) + 1.0 / UA_air)      # W/K

    m_dot_a_seg = rho_a * Va * (2.0 * Lf) * dz           # kg/s
    C_a_seg = m_dot_a_seg * cp_a                         # W/K
    NTU = UA_seg / C_a_seg
    eps_seg = 1.0 - np.exp(-NTU)

    q_seg = C_a_seg * (Tsat_K - Ta_in_K) * eps_seg       # W

    # Exit quality
    x_out = x_in - q_seg / q_chip
    if x_out < 0.0:
        x_out = 1.0e-5
        q_seg = (x_in - x_out) * q_chip
```

```
    # Exit void fraction
    alpha_out = 1.0 / (
        1.0
        + ((1.0 - x_out) / x_out)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Pressure drop model
    Gamma_in = G**2 * (
        x_in**2 / (rho_v * alpha_in)
        + (1.0 - x_in)**2 / (rho_l * (1.0 - alpha_in))
    )

    Gamma_out = G**2 * (
        x_out**2 / (rho_v * alpha_out)
        + (1.0 - x_out)**2 / (rho_l * (1.0 - alpha_out))
    )

    f_lo = 16.0 / Re_lo
    dPdz_lo = 2.0 * G**2 * f_lo / (D * rho_l)          # Pa/m

    if Re_vo < 2300.0:
        f_vo = 16.0 / Re_vo
    else:
        f_vo = 0.079 * Re_vo**(-0.25)

    rho_tp = 1.0 / (x_in / rho_v + (1.0 - x_in) / rho_l)
    Fr = G**2 / (rho_tp**2 * g * D)
    We = G**2 * D / (rho_tp * sigma)

    E = (1.0 - x_in)**2 + x_in**2 * (rho_l * f_vo) / (rho_v * f_lo)
    F = x_in**0.78 * (1.0 - x_in)**0.224
    H = (rho_l / rho_v)**0.91 * (mu_v / mu_l)**0.19 * (1.0 - mu_v / mu_l)**0.7

    phi_lo_sq = E + 3.24 * F * H / (Fr**0.045 * We**0.035)

    P_in_Pa = P_in_kPa * 1000.0
    P_out_Pa = P_in_Pa - phi_lo_sq * dPdz_lo * dz - (Gamma_out - Gamma_in)
    P_out_kPa = P_out_Pa / 1000.0

    return {
        "Tsat_C": props["Tsat_C"],          # °C
        "G": G,                             # kg/m^2·s
        "hr": hr,                           # W/m^2·K
        "UA_seg": UA_seg,                   # W/K
        "q_seg": q_seg,                     # W
        "x_out": x_out,                     # -
        "alpha_in": alpha_in,               # -
        "alpha_out": alpha_out,             # -
        "phi_lo_sq": phi_lo_sq,             # -
        "P_out_kPa": P_out_kPa              # kPa
    }

results = segment_model()

for key, value in results.items():
    if key == "Tsat_C":
        print(f"{key:12s} = {value:.6f} °C")
    elif key == "G":
        print(f"{key:12s} = {value:.6f} kg/m^2·s")
    elif key == "hr":
        print(f"{key:12s} = {value:.6f} W/m^2·K")
    elif key == "UA_seg":
        print(f"{key:12s} = {value:.6f} W/K")
    elif key == "q_seg":
        print(f"{key:12s} = {value:.6f} W")
    elif key == "P_out_kPa":
        print(f"{key:12s} = {value:.6f} kPa")
    else:
        print(f"{key:12s} = {value:.6f} -")
```

Output

Tsat_C = 42.603381 °C
G = 29.803984 kg/m^2·s
hr = 2731.501028 W/m^2·K
UA_seg = 2.313984 W/K
q_seg = 38.502144 W
x_out = 0.885758 -
alpha_in = 0.996895 -
alpha_out = 0.934745 -
phi_lo_sq = 26.055362 -
P_out_kPa = 219.981031 kPa

Appendix B - Task 2 Code

The code below consists of Part 1, Part 2(a), and Part 2(b) solutions for Task 2.

```
''' Task 2 '''

import numpy as np
import math
import matplotlib.pyplot as plt
from numpy.polynomial import polynomial as P

#define the five evaporator pressures (P2)
P2_vals = [160.0, 180.0, 200.0, 220.0, 240.0] #kPa

q_chip=340.0        # W
Ta_in_C=20.0        # C
Va=6.5              # m/s
dz=0.040          # m
D=0.003             # m
n_tubes=9           # -
rho_a=1.3           # kg/m^3
g=9.81               # m/s^2
segments_count = 20     #number of segments
A_chip = (6e-2)**2  #m^2
q_double_dot = q_chip/A_chip #W/m^2
T_design_chip = 55.0 + 273.15 #K
theta = 35 * np.pi/180.0 #radians

#from the image in the description of Task 2
x_s0= 0.999
alpha_s0 = 0.999

cp_a=1005.0         # J/kg·K
Lf=5.26e-3          # m

# Tsat(P2) constants
C0 = 20.472
C1 = -2579.9

#Lagrange interpolation for Thermophysical Properties

# ------------------------------
# R245fa saturation property data
# ------------------------------
T_data_C = np.array([15.1, 30.0, 46.0, 60.0])          #C

rho_l_data = np.array([1365.0, 1325.0, 1279.0, 1267.0])      # kg/m^3
rho_v_data = np.array([5.961, 10.169, 17.30, 25.68])         # kg/m^3
mu_l_data  = np.array([464e-6, 382e-6, 314e-6, 266e-6])      # Pa·s
mu_v_data  = np.array([9.93e-6, 10.4e-6, 11.0e-6, 11.5e-6])  # Pa·s
```

```
k_l_data   = np.array([0.0912, 0.0865, 0.0814, 0.0769])      # W/m·K
h_lv_data  = np.array([196.1e3, 187.0e3, 177.2e3, 168.0e3])  # J/kg0.
sigma_data = np.array([0.0153, 0.0134, 0.0114, 0.0096])      # N/m
cp_l_data  = np.array([1318.0, 1350.0, 1388.0, 1428.0])      # J/kg·K

def lagrange_interp(x, xdata, ydata):
    n = len(xdata)
    y = 0.0
    for i in range(n):
        Li = 1.0
        for j in range(n):
            if i != j:
                Li *= (x - xdata[j]) / (xdata[i] - xdata[j])
        y += ydata[i] * Li
    return y


def Tsat_from_P(P_kPa):
    P_Pa = P_kPa * 1000.0
    Tsat_K = C1 / (np.log(P_Pa) - C0)
    return Tsat_K


def r245fa_sat_props(P_kPa):
    Tsat_K = Tsat_from_P(P_kPa)
    Tsat_C = Tsat_K - 273.15

    rho_l = lagrange_interp(Tsat_C, T_data_C, rho_l_data)
    rho_v = lagrange_interp(Tsat_C, T_data_C, rho_v_data)
    mu_l  = lagrange_interp(Tsat_C, T_data_C, mu_l_data)
    mu_v  = lagrange_interp(Tsat_C, T_data_C, mu_v_data)
    k_l   = lagrange_interp(Tsat_C, T_data_C, k_l_data)
    h_lv  = lagrange_interp(Tsat_C, T_data_C, h_lv_data)
    sigma = lagrange_interp(Tsat_C, T_data_C, sigma_data)
    cp_l  = lagrange_interp(Tsat_C, T_data_C, cp_l_data)

    Pr_l = mu_l * cp_l / k_l

    return {
        "Tsat_K": Tsat_K,
        "Tsat_C": Tsat_C,
        "rho_l": rho_l,
        "rho_v": rho_v,
        "mu_l": mu_l,
        "mu_v": mu_v,
        "k_l": k_l,
        "h_lv": h_lv,
        "sigma": sigma,
        "cp_l": cp_l,
        "Pr_l": Pr_l
    }

units = {
    "Tsat_K": "K",
    "Tsat_C": "°C",
    "rho_l": "kg/m^3",
    "rho_v": "kg/m^3",
    "mu_l": "Pa·s",
    "mu_v": "Pa·s",
    "k_l": "W/m·K",
    "h_lv": "J/kg",
    "sigma": "N/m",
    "cp_l": "J/kg·K",
    "Pr_l": "-"
}


#repeated from Task 1 to define the segment calculating function:
#Single Segment Condenser Element

def segment_model(
    q_chip=340.0,       # W
```

```
    Ta_in_C=20.0,        # °C
    Va=6.5,              # m/s
    P_in_kPa=220.0,      # kPa
    dz=0.040,            # m
    D=0.003,             # m
    n_tubes=9,           # -
    x_in=0.999,          # -
    rho_a=1.3,           # kg/m^3
    cp_a=1005.0,         # J/kg·K
    Lf=5.26e-3,          # m
    g=9.81               # m/s^2
):
    props = r245fa_sat_props(P_in_kPa)

    Tsat_K = props["Tsat_K"]
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]
    k_l = props["k_l"]
    h_lv = props["h_lv"]
    sigma = props["sigma"]
    Pr_l = props["Pr_l"]

    Ta_in_K = Ta_in_C + 273.15

    # Mass flux
    A_passage = np.pi * D**2 / 4.0                        # m^2
    A_total = n_tubes * A_passage                         # m^2
    G = q_chip / (A_total * h_lv)                         # kg/m^2·s

    # Reynolds numbers
    Re_lo = G * D / mu_l
    Re_l = G * (1.0 - x_in) * D / mu_l
    Re_vo = G * D / mu_v

    # Martinelli parameter
    X_tt = ((1.0 - x_in) / x_in)**0.9 * (rho_v / rho_l)**0.5 * (mu_l / mu_v)**0.1

    # Traviss F_T
    if Re_l > 1125:
        F_T = 0.812 * (Pr_l**0.5) + 5.0 * np.log(1.0 + 5.0 * Pr_l) + 2.5 * np.log(0.0031 *
Re_l)
    elif Re_l > 50:
        F_T = 0.585 * (Pr_l**0.5) + 5.0 * np.log(1.0 + Pr_l * (0.0964 * Re_l**0.585 - 1.0))
    else:
        F_T = 0.5 * (Pr_l**0.5) * Re_l

    # Refrigerant-side heat transfer coefficient
    hr = (k_l / D) * 0.15 * (Pr_l**0.4) * (Re_l**0.9) * F_T * (1.0 / X_tt + 2.85 /
X_tt**0.476)

    # Inlet void fraction
    alpha_in = 1.0 / (
        1.0
        + ((1.0 - x_in) / x_in)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Segment UA and heat transfer
    Ar = n_tubes * np.pi * D * dz                          # m^2
    UA_air = 77.1 * (Va / 6.5)**0.5 * dz                  # W/K
    UA_seg = 1.0 / (1.0 / (hr * Ar) + 1.0 / UA_air)       # W/K

    m_dot_a_seg = rho_a * Va * (2.0 * Lf) * dz            # kg/s
    C_a_seg = m_dot_a_seg * cp_a                          # W/K
    NTU = UA_seg / C_a_seg
    eps_seg = 1.0 - np.exp(-NTU)

    q_seg = C_a_seg * (Tsat_K - Ta_in_K) * eps_seg        # W
```

```
    # Exit quality
    x_out = x_in - q_seg / q_chip
    if x_out < 0.0:
        x_out = 1.0e-5
        q_seg = (x_in - x_out) * q_chip

    # Exit void fraction
    alpha_out = 1.0 / (
        1.0
        + ((1.0 - x_out) / x_out)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Pressure drop model
    Gamma_in = G**2 * (
        x_in**2 / (rho_v * alpha_in)
        + (1.0 - x_in)**2 / (rho_l * (1.0 - alpha_in))
    )

    Gamma_out = G**2 * (
        x_out**2 / (rho_v * alpha_out)
        + (1.0 - x_out)**2 / (rho_l * (1.0 - alpha_out))
    )

    f_lo = 16.0 / Re_lo
    dPdz_lo = 2.0 * G**2 * f_lo / (D * rho_l)          # Pa/m

    if Re_vo < 2300.0:
        f_vo = 16.0 / Re_vo
    else:
        f_vo = 0.079 * Re_vo**(-0.25)

    rho_tp = 1.0 / (x_in / rho_v + (1.0 - x_in) / rho_l)
    Fr = G**2 / (rho_tp**2 * g * D)
    We = G**2 * D / (rho_tp * sigma)

    E = (1.0 - x_in)**2 + x_in**2 * (rho_l * f_vo) / (rho_v * f_lo)
    F = x_in**0.78 * (1.0 - x_in)**0.224
    H = (rho_l / rho_v)**0.91 * (mu_v / mu_l)**0.19 * (1.0 - mu_v / mu_l)**0.7

    phi_lo_sq = E + 3.24 * F * H / (Fr**0.045 * We**0.035)

    P_in_Pa = P_in_kPa * 1000.0
    P_out_Pa = P_in_Pa - phi_lo_sq * dPdz_lo * dz - (Gamma_out - Gamma_in)
    P_out_kPa = P_out_Pa / 1000.0

    return {
        "Tsat_C": props["Tsat_C"],          # °C
        "G": G,                             # kg/m^2·s
        "hr": hr,                           # W/m^2·K
        "UA_seg": UA_seg,                   # W/K
        "q_seg": q_seg,                     # W
        "x_out": x_out,                     # -
        "alpha_in": alpha_in,               # -
        "alpha_out": alpha_out,             # -
        "phi_lo_sq": phi_lo_sq,             # -
        "P_out_kPa": P_out_kPa              # kPa
    }

## Part 1
-------------------------------------------------------------------------------------------
-----------------------------

#find T_sat and T_w
def T_wall_calculator(q_chip, A_chip, T_chip_design_K, P_vals):
    '''
    Find T_w and the pressure that satisfies the chip temperature limit.
    '''
    T_chip_list = []
    for P_kPa in P_vals:
        props = r245fa_sat_props(P_kPa)
```

```
        Tsat  = props["Tsat_K"]
        rho_l = props["rho_l"]
        rho_v = props["rho_v"]
        k_l   = props["k_l"]
        Pr_l  = props["Pr_l"]
        sigma = props["sigma"]

        #to find T_w:

        Db = 0.0208 * theta * np.sqrt(sigma / (g * (rho_l - rho_v)))

        # B = (rho_v / rho_l)^0.581 * Pr_l^0.533
        # use the rearranged eq from notes: T_w =
[(1/207)(q_double_dot*Db/k_l)(k_l*T_sat/q_double_dot*Db)^0.745 *(B)^-1] + T_sat

        B = ((rho_v / rho_l)**0.581) * (Pr_l**0.533)
        T_wall = [(1/207)*(q_double_dot*Db/k_l)*((k_l*Tsat/q_double_dot*Db)**0.745)
*(B**(-1))] + Tsat

        T_chip_list.append(T_wall)

    return T_chip_list

#plot T_chip vs P_ev for Part 1 Deliverable
T_chip_list = T_wall_calculator(q_chip, A_chip, T_design_chip, P2_vals)
T_chip_C = [T - 273.15 for T in T_chip_list]

plt.figure(figsize=(8, 6))
plt.plot(P2_vals, T_chip_C, 'o-', label='Predicted T_chip')
plt.axhline(T_design_chip - 273.15, linestyle='--', label=f'Design limit
{T_design_chip-273.15:.0f} C')
plt.xlabel('Evaporator pressure P_2 [kPa]')
plt.ylabel('Chip temperature [C]')
plt.title('Chip Temperature vs Evaporator Pressure')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()


## Use a curve-fit to find the evaporator pressure that satisfies the specified q_dot at
T_chip = 55C
P_array = np.array(P2_vals)
T_array = np.array(T_chip_list).flatten()

#fit a first degree polynomial to T_chip vs P_2 -> P_2(T_chip) since T_chip is independent
coeffs = np.polyfit(T_array - 273.15, P_array, deg=1)
poly   = np.poly1d(coeffs)

#evaluate P at T_chip = 55°C
P_design = poly(55.0)

print(f"\n[Part 1] Design evaporator pressure: {P_design:.2f} kPa, chip will be at 55C")


#check the fit on the plot
T_plot = np.linspace(min(T_array - 273.15), 60.0, 200)
plt.figure(figsize=(7, 4))
plt.plot(P_array, T_array - 273.15, 'o', label='Computed points')
plt.plot(poly(T_plot), T_plot, 'r-', label='Polynomial fit')
plt.axhline(55.0, color='gray', linestyle='--', label='55C design limit')
plt.axvline(P_design, color='green', linestyle='--', label=f'P_design = {P_design:.1f}
kPa')
plt.xlabel('Evaporator pressure [kPa]')
plt.ylabel('Chip temperature [C]')
plt.title('Curve-fit: T_chip vs P_2')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
## Part 2
-----------------------------------------------------------------------------------------------
------------------------

#part 2(a) Loop and Sim Setup for Full Condenser
def condenser_sim(q_chip, Ta_in_C, Va, P_design, dz, D, n_tubes, segments_count, rho_a,
cp_a, Lf, g):
    '''
    Iterates through the 20 condenser segments, returning exit states of all segments and
the total
    properties of heat transfer rate and pressure drop and final exit state of the
condenser exit.
    '''
    #at inlet of s0: saturated vapour at evaporator pressure enters condenser
    x = x_s0
    P_kPa = P_design

    records = []
    q_total   = 0.0
    dP_total  = 0.0

    for i in range(segments_count):
        res = segment_model(
            q_chip  = q_chip, Ta_in_C = Ta_in_C, Va = Va,
            P_in_kPa= P_kPa, dz = dz, D = D, n_tubes = n_tubes, x_in = x,
            rho_a = rho_a, cp_a = cp_a, Lf = Lf, g = g
            )

        x_out = res["x_out"]
        alpha_out = res["alpha_out"]
        P_out_kPa = res["P_out_kPa"]
        q_seg = res["q_seg"]

        records.append({
            "seg"      : i + 1,
            "x_out"    : x_out,
            "alpha_out": alpha_out,
            "P_out_kPa": P_out_kPa,
            "q_seg"    : q_seg
        })

        #total rate of heat transfer calc
        q_total  += q_seg
        #total pressure drop calc
        dP_total += (P_kPa - P_out_kPa)

        #next segment state information
        x = x_out
        P_kPa = P_out_kPa

    return records, q_total, dP_total, P_kPa, x, alpha_out


#report the requested values for Part 2(a)
#export the requested values in Part 2(a) by calling the function defined above that houses
the segments for loop
records, q_total, dP_total, P_final_kPa, x_final, alpha_final = condenser_sim(q_chip,
Ta_in_C, Va, P_design, dz,
                                                                D, n_tubes,
segments_count, rho_a, cp_a, Lf, g)
print("\n[Part 2a] Condenser Full Length Results")
print(f"{'Seg':>4} {'x_out':>8} {'α_out':>8} {'P_out (kPa)':>10} {'q_seg (W)':>10}")

for r in records:
    print(f"{r['seg']:>4} {r['x_out']:>8.5f} {r['alpha_out']:>8.5f} {r['P_out_kPa']:>12.4f}
{r['q_seg']:>10.4f}")

print(f"\nTotal refrigerant pressure drop: {dP_total:.4f} kPa")
print(f"Total condenser heat transfer: {q_total:.2f} W")
print(f"Condenser exit pressure: {P_final_kPa:.4f} kPa")
print(f"Condenser exit quality: {x_final:.6f}")
print(f"Condenser exit void fraction: {alpha_final:.6f}")
```

```
if x_final <= 0.0001:
    print("Exit state: saturated liquid -> full condensation achieved")
else:
    print("Exit state: two-phase mixture")


#Part 2(b) Find the total length of the condenser for x = 0 (qulaity to be equal to zero
exactly)
def best_length_search(q_chip, Ta_in_C, Va, P_design, dz, D, n_tubes,
                                  rho_a, cp_a, Lf, g, segments_count):
    '''
    Finds the total condenser length for which x becomes zero.
    '''

    #initialize a total length and define properties
    L = 0.0
    P_kPa = P_design
    x = x_s0

    #set up the loop to search
    for seg in range(segments_count):
        if x <= 0.0001:
            break
        sols = segment_model(q_chip=q_chip, Ta_in_C=Ta_in_C, Va=Va,
                             P_in_kPa=P_kPa, dz=dz, D=D, n_tubes=n_tubes,
                             x_in=x, rho_a=rho_a, cp_a=cp_a, Lf=Lf, g=g
                             )
        L += dz
        P_kPa = sols["P_out_kPa"]
        x = sols["x_out"]

    return L

#export the total length that achieves a quality of zero by calling the function
L_x_0 = best_length_search(q_chip, Ta_in_C, Va,
                           P_design, dz, D, n_tubes,
                           rho_a, cp_a, Lf, g, segments_count)

#report the final length val for Part 2(b)
print(f"\n[Part 2b] The total condenser length that achieves a quality of zero at the exit
is: {L_x_0:.3f} m" )
```

Appendix C - Task 3 Code

#Appendix C1 - Modified Condenser Geometry and Performance Evaluation

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# ============================================================
# GIVEN / BASE CONDITIONS
# ============================================================

q_chip = 340.0          # W
Ta_in_C = 20.0          # C
Va = 6.5                # m/s
dz = 0.040              # m
segments_count = 20
rho_a = 1.3             # kg/m^3
cp_a = 1005.0           # J/kg-K
Lf = 5.26e-3            # m
g = 9.81                # m/s^2

A_chip = (6e-2)**2      # m^2
theta = 35 * np.pi / 180.0
q_double_dot = q_chip / A_chip
T_design_chip = 55.0 + 273.15   # K
```

```
# inlet state to condenser
x_s0 = 0.999
alpha_s0 = 0.999

P_design = 285.63   # kPa  # Task 2 design evaporator pressure


# ============================================================
# SATURATION PROPERTY MODEL
# ============================================================

C0 = 20.472
C1 = -2579.9

T_data_C = np.array([15.1, 30.0, 46.0, 60.0])
rho_l_data = np.array([1365.0, 1325.0, 1279.0, 1267.0])      # kg/m^3
rho_v_data = np.array([5.961, 10.169, 17.30, 25.68])         # kg/m^3
mu_l_data  = np.array([464e-6, 382e-6, 314e-6, 266e-6])      # Pa·s
mu_v_data  = np.array([9.93e-6, 10.4e-6, 11.0e-6, 11.5e-6])  # Pa·s
k_l_data   = np.array([0.0912, 0.0865, 0.0814, 0.0769])      # W/m·K
h_lv_data  = np.array([196.1e3, 187.0e3, 177.2e3, 168.0e3])  # J/kg
sigma_data = np.array([0.0153, 0.0134, 0.0114, 0.0096])      # N/m
cp_l_data  = np.array([1318.0, 1350.0, 1388.0, 1428.0])      # J/kg·K

def lagrange_interp(x, xdata, ydata):
   n = len(xdata)
   y = 0.0
   for i in range(n):
       Li = 1.0
       for j in range(n):
           if i != j:
               Li *= (x - xdata[j]) / (xdata[i] - xdata[j])
       y += ydata[i] * Li
   return y

def Tsat_from_P(P_kPa):
   P_Pa = P_kPa * 1000.0
   return C1 / (np.log(P_Pa) - C0)

def r245fa_sat_props(P_kPa):
   Tsat_K = Tsat_from_P(P_kPa)
   Tsat_C = Tsat_K - 273.15

   rho_l = lagrange_interp(Tsat_C, T_data_C, rho_l_data)
   rho_v = lagrange_interp(Tsat_C, T_data_C, rho_v_data)
   mu_l  = lagrange_interp(Tsat_C, T_data_C, mu_l_data)
   mu_v  = lagrange_interp(Tsat_C, T_data_C, mu_v_data)
   k_l   = lagrange_interp(Tsat_C, T_data_C, k_l_data)
   h_lv  = lagrange_interp(Tsat_C, T_data_C, h_lv_data)
   sigma = lagrange_interp(Tsat_C, T_data_C, sigma_data)
   cp_l  = lagrange_interp(Tsat_C, T_data_C, cp_l_data)

   Pr_l = mu_l * cp_l / k_l

   return {
       "Tsat_K": Tsat_K,
       "Tsat_C": Tsat_C,
       "rho_l": rho_l,
       "rho_v": rho_v,
       "mu_l": mu_l,
       "mu_v": mu_v,
       "k_l": k_l,
       "h_lv": h_lv,
       "sigma": sigma,
       "cp_l": cp_l,
       "Pr_l": Pr_l
   }

# ============================================================
# SEGMENT MODEL
# ============================================================
```

```
def segment_model(
    q_chip=340.0,
    Ta_in_C=20.0,
    Va=6.5,
    P_in_kPa=285.6,
    dz=0.040,
    D=0.003,
    n_tubes=9,
    x_in=0.999,
    rho_a=1.3,
    cp_a=1005.0,
    Lf=5.26e-3,
    g=9.81
):
    props = r245fa_sat_props(P_in_kPa)

    Tsat_K = props["Tsat_K"]
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]
    k_l = props["k_l"]
    h_lv = props["h_lv"]
    sigma = props["sigma"]
    Pr_l = props["Pr_l"]

    Ta_in_K = Ta_in_C + 273.15

    # Flow area and mass flux
    A_passage = np.pi * D**2 / 4.0
    A_total = n_tubes * A_passage
    G = q_chip / (A_total * h_lv)

    # Reynolds numbers
    Re_lo = G * D / mu_l
    Re_l = G * (1.0 - x_in) * D / mu_l
    Re_vo = G * D / mu_v

    # Guard against zero / invalid values
    x_safe = max(x_in, 1.0e-6)
    one_minus_x_safe = max(1.0 - x_in, 1.0e-6)
    Re_l_safe = max(Re_l, 1.0e-6)

    # Martinelli parameter
    X_tt = ((one_minus_x_safe / x_safe)**0.9) * (rho_v / rho_l)**0.5 * (mu_l / mu_v)**0.1
    X_tt = max(X_tt, 1.0e-8)

    # Traviss factor
    if Re_l > 1125:
        F_T = 0.812 * (Pr_l**0.5) + 5.0 * np.log(1.0 + 5.0 * Pr_l) + 2.5 * np.log(0.0031 *
Re_l_safe)
    elif Re_l > 50:
        F_T = 0.585 * (Pr_l**0.5) + 5.0 * np.log(1.0 + Pr_l * (0.0964 * Re_l_safe**0.585 -
1.0))
    else:
        F_T = 0.5 * (Pr_l**0.5) * Re_l_safe

    # Refrigerant-side h
    hr = (k_l / D) * 0.15 * (Pr_l**0.4) * (Re_l_safe**0.9) * F_T * (1.0 / X_tt + 2.85 /
X_tt**0.476)

    # Void fraction at inlet
    alpha_in = 1.0 / (
        1.0
        + ((one_minus_x_safe / x_safe)**0.64)
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # UA and segment heat transfer
    Ar = n_tubes * np.pi * D * dz
    UA_air = 77.1 * (Va / 6.5)**0.5 * dz
```

```
    UA_seg = 1.0 / (1.0 / (hr * Ar) + 1.0 / UA_air)

    m_dot_a_seg = rho_a * Va * (2.0 * Lf) * dz
    C_a_seg = m_dot_a_seg * cp_a
    NTU = UA_seg / C_a_seg
    eps_seg = 1.0 - np.exp(-NTU)

    q_seg = C_a_seg * (Tsat_K - Ta_in_K) * eps_seg

    # Exit quality
    x_out = x_in - q_seg / q_chip
    if x_out < 0.0:
        x_out = 1.0e-5
        q_seg = (x_in - x_out) * q_chip

    x_out_safe = max(x_out, 1.0e-6)
    one_minus_x_out_safe = max(1.0 - x_out, 1.0e-6)

    # Exit void fraction
    alpha_out = 1.0 / (
        1.0
        + ((one_minus_x_out_safe / x_out_safe)**0.64)
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Pressure drop model
    Gamma_in = G**2 * (
        x_safe**2 / (rho_v * alpha_in)
        + one_minus_x_safe**2 / (rho_l * (1.0 - alpha_in))
    )

    Gamma_out = G**2 * (
        x_out_safe**2 / (rho_v * alpha_out)
        + one_minus_x_out_safe**2 / (rho_l * (1.0 - alpha_out))
    )

    f_lo = 16.0 / max(Re_lo, 1.0e-6)
    dPdz_lo = 2.0 * G**2 * f_lo / (D * rho_l)

    if Re_vo < 2300.0:
        f_vo = 16.0 / max(Re_vo, 1.0e-6)
    else:
        f_vo = 0.079 * Re_vo**(-0.25)

    rho_tp = 1.0 / (x_safe / rho_v + one_minus_x_safe / rho_l)
    Fr = G**2 / (rho_tp**2 * g * D)
    We = G**2 * D / (rho_tp * sigma)

    E = one_minus_x_safe**2 + x_safe**2 * (rho_l * f_vo) / (rho_v * f_lo)
    F = x_safe**0.78 * one_minus_x_safe**0.224
    H = (rho_l / rho_v)**0.91 * (mu_v / mu_l)**0.19 * (1.0 - mu_v / mu_l)**0.7

    phi_lo_sq = E + 3.24 * F * H / (Fr**0.045 * We**0.035)

    P_out_Pa = P_in_kPa * 1000.0 - phi_lo_sq * dPdz_lo * dz - (Gamma_out - Gamma_in)
    P_out_kPa = P_out_Pa / 1000.0

    return {
        "Tsat_C": props["Tsat_C"],
        "G": G,
        "hr": hr,
        "UA_seg": UA_seg,
        "q_seg": q_seg,
        "x_out": x_out,
        "alpha_in": alpha_in,
        "alpha_out": alpha_out,
        "phi_lo_sq": phi_lo_sq,
        "P_out_kPa": P_out_kPa
    }

# ============================================================
```

```python
# CONDENSER LOOP MODEL
# ============================================================

def condenser_sim(q_chip, Ta_in_C, Va, P_design, dz, D, n_tubes, segments_count, rho_a,
cp_a, Lf, g):
    x = x_s0
    P_kPa = P_design

    records = []
    q_total = 0.0

    for i in range(segments_count):
        res = segment_model(
            q_chip=q_chip,
            Ta_in_C=Ta_in_C,
            Va=Va,
            P_in_kPa=P_kPa,
            dz=dz,
            D=D,
            n_tubes=n_tubes,
            x_in=x,
            rho_a=rho_a,
            cp_a=cp_a,
            Lf=Lf,
            g=g
        )

        record = {
            "Segment": i + 1,
            "x_out": res["x_out"],
            "alpha_out": res["alpha_out"],
            "P_out_kPa": res["P_out_kPa"],
            "q_seg_W": res["q_seg"],
            "hr_W_m2K": res["hr"],
            "G_kg_m2s": res["G"]
        }
        records.append(record)

        x = res["x_out"]
        P_kPa = res["P_out_kPa"]
        q_total += res["q_seg"]

    dP_total = P_design - P_kPa

    return {
        "records": pd.DataFrame(records),
        "q_total_W": q_total,
        "dP_total_kPa": dP_total,
        "P_exit_kPa": P_kPa,
        "x_exit": x,
        "alpha_exit": records[-1]["alpha_out"]
    }

# ============================================================
# RUN BASE CASE (TASK 2) AND NEW CASE (TASK 3a)
# ============================================================

# Original Task 2 geometry
D_old = 0.0030
n_tubes_old = 9

# New Task 3(a) geometry
D_new = 0.0024
n_tubes_new = 10

old_case = condenser_sim(
    q_chip, Ta_in_C, Va, P_design, dz, D_old, n_tubes_old,
    segments_count, rho_a, cp_a, Lf, g
)

new_case = condenser_sim(
    q_chip, Ta_in_C, Va, P_design, dz, D_new, n_tubes_new,
```

```
    segments_count, rho_a, cp_a, Lf, g
)

# ============================================================
# BUILD COMPARISON TABLE
# ============================================================

df_old = old_case["records"].copy()
df_new = new_case["records"].copy()

comparison = pd.DataFrame({
    "Segment": df_old["Segment"],
    "x_out_old": df_old["x_out"],
    "x_out_new": df_new["x_out"],
    "alpha_out_old": df_old["alpha_out"],
    "alpha_out_new": df_new["alpha_out"],
    "P_out_old_kPa": df_old["P_out_kPa"],
    "P_out_new_kPa": df_new["P_out_kPa"],
    "q_seg_old_W": df_old["q_seg_W"],
    "q_seg_new_W": df_new["q_seg_W"]
})

pd.set_option("display.max_rows", None)
pd.set_option("display.float_format", lambda x: f"{x:0.5f}")

print("\n==================== TASK 3(a): SEGMENT-BY-SEGMENT COMPARISON
====================\n")
print(comparison.to_string(index=False))

# ============================================================
# OVERALL RESULTS
# ============================================================

print("\n==================== OVERALL CONDENSER RESULTS ====================\n")

print("Original Task 2 geometry: 9 passages, D = 3.0 mm")
print(f"  Total condenser heat transfer = {old_case['q_total_W']:.3f} W")
print(f"  Total condenser pressure drop = {old_case['dP_total_kPa']:.5f} kPa")
print(f"  Condenser exit pressure       = {old_case['P_exit_kPa']:.5f} kPa")
print(f"  Condenser exit quality        = {old_case['x_exit']:.6f}")
print(f"  Condenser exit void fraction  = {old_case['alpha_exit']:.6f}\n")

print("New Task 3(a) geometry: 10 passages, D = 2.4 mm")
print(f"  Total condenser heat transfer = {new_case['q_total_W']:.3f} W")
print(f"  Total condenser pressure drop = {new_case['dP_total_kPa']:.5f} kPa")
print(f"  Condenser exit pressure       = {new_case['P_exit_kPa']:.5f} kPa")
print(f"  Condenser exit quality        = {new_case['x_exit']:.6f}")
print(f"  Condenser exit void fraction  = {new_case['alpha_exit']:.6f}\n")

print("==================== DIFFERENCE: NEW - OLD ====================\n")
print(f"Delta total heat transfer = {new_case['q_total_W'] - old_case['q_total_W']:.3f} W")
print(f"Delta total pressure drop = {new_case['dP_total_kPa'] -
old_case['dP_total_kPa']:.5f} kPa")
print(f"Delta exit quality        = {new_case['x_exit'] - old_case['x_exit']:.6f}")
print(f"Delta exit pressure       = {new_case['P_exit_kPa'] - old_case['P_exit_kPa']:.5f}
kPa")

# ============================================================
# OPTIONAL PLOTS FOR REPORT
# ============================================================

plt.figure(figsize=(7, 5))
plt.plot(comparison["Segment"], comparison["x_out_old"], 'o-', label="Old: 9 x 3.0 mm")
plt.plot(comparison["Segment"], comparison["x_out_new"], 's-', label="New: 10 x 2.4 mm")
plt.xlabel("Segment")
plt.ylabel("Exit quality")
plt.title("Segment exit quality comparison")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(7, 5))
plt.plot(comparison["Segment"], comparison["alpha_out_old"], 'o-', label="Old: 9 x 3.0 mm")
plt.plot(comparison["Segment"], comparison["alpha_out_new"], 's-', label="New: 10 x 2.4
mm")
plt.xlabel("Segment")
plt.ylabel("Exit void fraction")
plt.title("Segment exit void fraction comparison")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

plt.figure(figsize=(7, 5))
plt.plot(comparison["Segment"], comparison["P_out_old_kPa"], 'o-', label="Old: 9 x 3.0 mm")
plt.plot(comparison["Segment"], comparison["P_out_new_kPa"], 's-', label="New: 10 x 2.4
mm")
plt.xlabel("Segment")
plt.ylabel("Exit pressure [kPa]")
plt.title("Segment exit pressure comparison")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

Output

```
==================== TASK 3(a): SEGMENT-BY-SEGMENT COMPARISON ====================
```

Segment	x_out_old	x_out_new	alpha_out_old	alpha_out_new	P_out_old_kPa	P_out_new_kPa	q_seg_old_W	q_seg_new_W
1	0.83168	0.82342	0.90611	0.90292	285.61926	285.60453	56.88962	59.69570
2	0.64013	0.62552	0.83386	0.82821	285.59266	285.54650	65.12573	67.28891
3	0.44096	0.42600	0.74891	0.74152	285.56702	285.49141	67.71724	67.83663
4	0.24165	0.22645	0.62546	0.61268	285.54483	285.44472	67.76659	67.84490
5	0.04257	0.02715	0.32132	0.26007	285.52819	285.41067	67.68828	67.76294
6	0.00001	0.00001	0.00219	0.00219	285.52137	285.39948	14.46915	9.22752
7	0.00001	0.00001	0.00219	0.00219	285.52036	285.39725	0.00000	0.00000
8	0.00001	0.00001	0.00219	0.00219	285.51934	285.39503	0.00000	0.00000
9	0.00001	0.00001	0.00219	0.00219	285.51833	285.39280	0.00000	0.00000
10	0.00001	0.00001	0.00219	0.00219	285.51732	285.39057	0.00000	0.00000
11	0.00001	0.00001	0.00219	0.00219	285.51630	285.38834	0.00000	0.00000
12	0.00001	0.00001	0.00219	0.00219	285.51529	285.38612	0.00000	0.00000
13	0.00001	0.00001	0.00219	0.00219	285.51427	285.38389	0.00000	0.00000
14	0.00001	0.00001	0.00219	0.00219	285.51326	285.38166	0.00000	0.00000
15	0.00001	0.00001	0.00219	0.00219	285.51224	285.37943	0.00000	0.00000
16	0.00001	0.00001	0.00219	0.00219	285.51123	285.37721	0.00000	0.00000
17	0.00001	0.00001	0.00219	0.00219	285.51022	285.37498	0.00000	0.00000
18	0.00001	0.00001	0.00219	0.00219	285.50920	285.37275	0.00000	0.00000
19	0.00001	0.00001	0.00219	0.00219	285.50819	285.37052	0.00000	0.00000
20	0.00001	0.00001	0.00219	0.00219	285.50717	285.36830	0.00000	0.00000

```
==================== OVERALL CONDENSER RESULTS ====================

Original Task 2 geometry: 9 passages, D = 3.0 mm
  Total condenser heat transfer = 339.657 W
  Total condenser pressure drop = 0.12283 kPa
  Condenser exit pressure       = 285.50717 kPa
  Condenser exit quality        = 0.000010
  Condenser exit void fraction  = 0.002186

New Task 3(a) geometry: 10 passages, D = 2.4 mm
  Total condenser heat transfer = 339.657 W
  Total condenser pressure drop = 0.26170 kPa
  Condenser exit pressure       = 285.36830 kPa
  Condenser exit quality        = 0.000010
  Condenser exit void fraction  = 0.002186

==================== DIFFERENCE: NEW - OLD ====================

Delta total heat transfer = 0.000 W
Delta total pressure drop = 0.13888 kPa
Delta exit quality        = 0.000000
Delta exit pressure       = -0.13888 kPa
```

#Appendix C2 - Determination of Required Condenser Length for Complete Condensation

```
def segment_model_no_clip(
    q_chip=340.0,
    Ta_in_C=20.0,
    Va=6.5,
    P_in_kPa=285.63,
    dz=0.040,
    D=0.0024,
    n_tubes=10,
    x_in=0.999,
    rho_a=1.3,
    cp_a=1005.0,
    Lf=5.26e-3,
    g=9.81
):

    props = r245fa_sat_props(P_in_kPa)

    Tsat_K = props["Tsat_K"]
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]
    k_l = props["k_l"]
    h_lv = props["h_lv"]
    sigma = props["sigma"]
    Pr_l = props["Pr_l"]

    Ta_in_K = Ta_in_C + 273.15

    # Flow area and mass flux
    A_passage = np.pi * D**2 / 4.0
    A_total = n_tubes * A_passage
    G = q_chip / (A_total * h_lv)

    # Reynolds numbers
    Re_lo = G * D / mu_l
    Re_l = G * max(1.0 - x_in, 1.0e-8) * D / mu_l
    Re_vo = G * D / mu_v

    x_safe = max(x_in, 1.0e-8)
    one_minus_x_safe = max(1.0 - x_in, 1.0e-8)

    # Martinelli parameter
    X_tt = ((one_minus_x_safe / x_safe)**0.9) * (rho_v / rho_l)**0.5 * (mu_l / mu_v)**0.1
    X_tt = max(X_tt, 1.0e-8)

    # Traviss factor
    if Re_l > 1125:
        F_T = 0.812 * (Pr_l**0.5) + 5.0 * np.log(1.0 + 5.0 * Pr_l) + 2.5 * np.log(0.0031 *
Re_l)
    elif Re_l > 50:
        F_T = 0.585 * (Pr_l**0.5) + 5.0 * np.log(1.0 + Pr_l * (0.0964 * Re_l**0.585 - 1.0))
    else:
        F_T = 0.5 * (Pr_l**0.5) * Re_l

    # Refrigerant-side heat transfer coefficient
    hr = (k_l / D) * 0.15 * (Pr_l**0.4) * (Re_l**0.9) * F_T * (1.0 / X_tt + 2.85 /
X_tt**0.476)

    # Inlet void fraction
    alpha_in = 1.0 / (
        1.0
        + ((one_minus_x_safe / x_safe)**0.64)
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # UA and heat transfer
    Ar = n_tubes * np.pi * D * dz
    UA_air = 77.1 * (Va / 6.5)**0.5 * dz
    UA_seg = 1.0 / (1.0 / (hr * Ar) + 1.0 / UA_air)
```

```
m_dot_a_seg = rho_a * Va * (2.0 * Lf) * dz
C_a_seg = m_dot_a_seg * cp_a
NTU = UA_seg / C_a_seg
eps_seg = 1.0 - np.exp(-NTU)

q_seg = C_a_seg * (Tsat_K - Ta_in_K) * eps_seg

# raw exit quality before clipping
x_out_raw = x_in - q_seg / q_chip

# clipped value only for pressure-drop calculation stability
if x_out_raw <= 0.0:
    x_out_clip = 1.0e-8
    q_seg_effective = x_in * q_chip   # only enough to reach x = 0
else:
    x_out_clip = x_out_raw
    q_seg_effective = q_seg

# Exit void fraction
x_out_safe = max(x_out_clip, 1.0e-8)
one_minus_x_out_safe = max(1.0 - x_out_clip, 1.0e-8)

alpha_out = 1.0 / (
    1.0
    + ((one_minus_x_out_safe / x_out_safe)**0.64)
    * (rho_v / rho_l)**0.36
    * (mu_l / mu_v)**0.07
)

# Pressure drop model
Gamma_in = G**2 * (
    x_safe**2 / (rho_v * alpha_in)
    + one_minus_x_safe**2 / (rho_l * (1.0 - alpha_in))
)

Gamma_out = G**2 * (
    x_out_safe**2 / (rho_v * alpha_out)
    + one_minus_x_out_safe**2 / (rho_l * (1.0 - alpha_out))
)

f_lo = 16.0 / max(Re_lo, 1.0e-8)
dPdz_lo = 2.0 * G**2 * f_lo / (D * rho_l)

if Re_vo < 2300.0:
    f_vo = 16.0 / max(Re_vo, 1.0e-8)
else:
    f_vo = 0.079 * Re_vo**(-0.25)

rho_tp = 1.0 / (x_safe / rho_v + one_minus_x_safe / rho_l)
Fr = G**2 / (rho_tp**2 * g * D)
We = G**2 * D / (rho_tp * sigma)

E = one_minus_x_safe**2 + x_safe**2 * (rho_l * f_vo) / (rho_v * f_lo)
F = x_safe**0.78 * one_minus_x_safe**0.224
H = (rho_l / rho_v)**0.91 * (mu_v / mu_l)**0.19 * (1.0 - mu_v / mu_l)**0.7

phi_lo_sq = E + 3.24 * F * H / (Fr**0.045 * We**0.035)

P_out_Pa = P_in_kPa * 1000.0 - phi_lo_sq * dPdz_lo * dz - (Gamma_out - Gamma_in)
P_out_kPa = P_out_Pa / 1000.0

return {
    "q_seg": q_seg,
    "q_seg_effective": q_seg_effective,
    "x_out_raw": x_out_raw,
    "x_out_clip": x_out_clip,
    "alpha_out": alpha_out,
    "P_out_kPa": P_out_kPa,
    "UA_seg": UA_seg,
    "hr": hr
}
```

```
def find_required_condenser_length(
    P_design = 285.63,   # kPa  # Task 2 design evaporator pressure
    q_chip=340.0,
    Ta_in_C=20.0,
    Va=6.5,
    dz_base=0.040,
    D=0.0024,
    n_tubes=10,
    rho_a=1.3,
    cp_a=1005.0,
    Lf=5.26e-3,
    g=9.81,
    x_start=0.999,
    max_segments=200
):
    """
    Finds the tube length required so that x reaches exactly 0 at the outlet.
    Uses linear interpolation within the crossing segment.
    """
    x = x_start
    P_kPa = P_design
    total_length = 0.0
    total_heat = 0.0
    records = []

    for i in range(max_segments):
        res = segment_model_no_clip(
            q_chip=q_chip,
            Ta_in_C=Ta_in_C,
            Va=Va,
            P_in_kPa=P_kPa,
            dz=dz_base,
            D=D,
            n_tubes=n_tubes,
            x_in=x,
            rho_a=rho_a,
            cp_a=cp_a,
            Lf=Lf,
            g=g
        )

        x_out_raw = res["x_out_raw"]

        # Case 1: condensation finishes inside this segment
        if x_out_raw <= 0.0:
            # fraction of this segment needed to go from x_in to x=0
            frac = x / (x - x_out_raw)
            dz_needed = frac * dz_base
            total_length += dz_needed

            # heat needed to reach x = 0 exactly
            q_needed = x * q_chip
            total_heat += q_needed

            records.append({
                "segment": i + 1,
                "x_in": x,
                "x_out_est": 0.0,
                "P_in_kPa": P_kPa,
                "P_out_kPa_est": np.nan,   # pressure at interpolated partial segment not
estimated exactly
                "dz_used_m": dz_needed,
                "crossed_zero": True
            })

            return {
                "required_length_m": total_length,
                "required_length_cm": total_length * 100.0,
                "segments_used": i + frac,
                "full_segments_completed": i,
```

```
                "heat_removed_W": total_heat,
                "records": pd.DataFrame(records)
            }

        # Case 2: still two-phase after full segment
        total_length += dz_base
        total_heat += res["q_seg"]

        records.append({
            "segment": i + 1,
            "x_in": x,
            "x_out_est": x_out_raw,
            "P_in_kPa": P_kPa,
            "P_out_kPa_est": res["P_out_kPa"],
            "dz_used_m": dz_base,
            "crossed_zero": False
        })

        x = x_out_raw
        P_kPa = res["P_out_kPa"]

    raise RuntimeError("Did not reach x = 0 within max_segments. Increase max_segments.")


# ============================================================
# RUN TASK 3(b)
# ============================================================

P_design = 285.63   # kPa

result_b = find_required_condenser_length(
    P_design=P_design,
    q_chip=340.0,
    Ta_in_C=20.0,
    Va=6.5,
    dz_base=0.040,
    D=0.0024,
    n_tubes=10,
    rho_a=1.3,
    cp_a=1005.0,
    Lf=5.26e-3,
    g=9.81,
    x_start=0.999
)

print("\n================ TASK 3(b) RESULT ================\n")
print(f"Required condenser tube length for x_out = 0: {result_b['required_length_m']:.6f}
m")
print(f"Required condenser tube length for x_out = 0: {result_b['required_length_cm']:.3f}
cm")
print(f"Equivalent number of 0.040 m segments: {result_b['segments_used']:.3f}")

print("\nSegment history up to condensation completion:\n")
print(result_b["records"].to_string(index=False))
```

Output

```
================ TASK 3(b) RESULT ================

Required condenser tube length for x_out = 0: 0.205504 m
Required condenser tube length for x_out = 0: 20.550 cm
Equivalent number of 0.040 m segments: 5.138

Segment history up to condensation completion:

 segment    x_in  x_out_est  P_in_kPa  P_out_kPa_est  dz_used_m  crossed_zero
       1 0.99900    0.82342 285.63000      285.60453    0.04000         False
       2 0.82342    0.62552 285.60453      285.54650    0.04000         False
       3 0.62552    0.42600 285.54650      285.49141    0.04000         False
       4 0.42600    0.22645 285.49141      285.44472    0.04000         False
       5 0.22645    0.02715 285.44472      285.41067    0.04000         False
       6 0.02715    0.00000 285.41067            NaN    0.00550          True
```

#Appendix C3 - System Performance Adjustment for Increased Inlet Air Temperature

```
# ----------------------------
# USER INPUTS
# ----------------------------
P_task2 = 285.63,
L_task3b = 0.205504,             # <-- Task 3(b) result [m]
Ta_in_new_C = 25.0              # new inlet air temperature for Task 3(c)

# fixed condenser geometry from Task 3(a)/(b)
D_new = 0.0024                  # m
n_tubes_new = 10

# other given conditions
q_chip = 340.0                  # W
Va_base = 6.5                   # m/s
rho_a = 1.3                     # kg/m^3
cp_a = 1005.0                   # J/kg-K
Lf = 5.26e-3                    # m
g = 9.81                        # m/s^2
dz_base = 0.040                 # m
x_s0 = 0.999                    # inlet quality to condenser
A_chip = (6e-2)**2              # m^2
theta = 35 * np.pi / 180.0
q_double_dot = q_chip / A_chip
T_chip_target_C = 55.0

# ------------------------------------------------------------
# CHIP TEMPERATURE MODEL (from Task 1 )
# ------------------------------------------------------------
def chip_temperature_from_pressure(P_kPa):
   """
   Returns predicted chip/wall temperature in deg C
   using the same boiling-side relation used in the earlier code.
   """
   props = r245fa_sat_props(P_kPa)

   Tsat = props["Tsat_K"]
   rho_l = props["rho_l"]
   rho_v = props["rho_v"]
   k_l = props["k_l"]
   Pr_l = props["Pr_l"]
   sigma = props["sigma"]

   Db = 0.0208 * theta * np.sqrt(sigma / (g * (rho_l - rho_v)))
   B = ((rho_v / rho_l)**0.581) * (Pr_l**0.533)

   # same expression structure as your Task 2 code
   T_wall_K = (1/207) * (q_double_dot * Db / k_l) * ((k_l * Tsat / (q_double_dot *
Db))**0.745) * (B**(-1)) + Tsat
   T_wall_C = T_wall_K - 273.15

   return T_wall_C


# ------------------------------------------------------------
# CONDENSER SIMULATION FOR ARBITRARY TOTAL LENGTH
# Allows final partial segment
# ------------------------------------------------------------
def condenser_sim_variable_length(
   total_length_m,
   q_chip=340.0,
   Ta_in_C=25.0,
   Va=6.5,
   P_design=285.63,
   dz_base=0.040,
   D=0.0024,
   n_tubes=10,
```

```
    rho_a=1.3,
    cp_a=1005.0,
    Lf=5.26e-3,
    g=9.81,
    x_start=0.999
):
    x = x_start
    P_kPa = P_design
    q_total = 0.0
    z_total = 0.0
    records = []

    n_full = int(total_length_m // dz_base)
    dz_last = total_length_m - n_full * dz_base

    segment_lengths = [dz_base] * n_full
    if dz_last > 1e-12:
        segment_lengths.append(dz_last)

    for i, dz_i in enumerate(segment_lengths):
        res = segment_model_no_clip(
            q_chip=q_chip,
            Ta_in_C=Ta_in_C,
            Va=Va,
            P_in_kPa=P_kPa,
            dz=dz_i,
            D=D,
            n_tubes=n_tubes,
            x_in=x,
            rho_a=rho_a,
            cp_a=cp_a,
            Lf=Lf,
            g=g
        )

        x_out_raw = res["x_out_raw"]

        # if condensation completes inside this segment
        if x_out_raw <= 0.0:
            frac = x / (x - x_out_raw)
            dz_used = frac * dz_i
            q_needed = x * q_chip

            z_total += dz_used
            q_total += q_needed

            records.append({
                "segment": i + 1,
                "dz_m": dz_used,
                "x_in": x,
                "x_out": 0.0,
                "P_in_kPa": P_kPa,
                "P_out_kPa": np.nan,
                "crossed_zero": True
            })

            return {
                "records": pd.DataFrame(records),
                "q_total_W": q_total,
                "P_exit_kPa": P_kPa,        # approximate, since final partial-segment P not
interpolated
                "x_exit": 0.0,
                "alpha_exit": 0.0,
                "used_length_m": z_total,
                "fully_condensed": True
            }

        # otherwise complete full/partial segment normally
        z_total += dz_i
        q_total += res["q_seg"]

        records.append({
```

```
            "segment": i + 1,
            "dz_m": dz_i,
            "x_in": x,
            "x_out": x_out_raw,
            "P_in_kPa": P_kPa,
            "P_out_kPa": res["P_out_kPa"],
            "crossed_zero": False
        })

        x = x_out_raw
        P_kPa = res["P_out_kPa"]

    # did not fully condense within total_length_m
    # estimate exit void fraction using small-quality clipping
    x_safe = max(x, 1.0e-8)
    props = r245fa_sat_props(P_kPa)
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]

    alpha_exit = 1.0 / (
        1.0
        + ((max(1.0 - x_safe, 1.0e-8) / x_safe)**0.64)
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    return {
        "records": pd.DataFrame(records),
        "q_total_W": q_total,
        "P_exit_kPa": P_kPa,
        "x_exit": x,
        "alpha_exit": alpha_exit,
        "used_length_m": z_total,
        "fully_condensed": False
    }


# ------------------------------------------------------------
# BASELINE TASK 3 SYSTEM AT 25 C
# ------------------------------------------------------------
baseline_case = condenser_sim_variable_length(
    total_length_m=L_task3b,
    q_chip=q_chip,
    Ta_in_C=Ta_in_new_C,
    Va=Va_base,
    P_design=P_task2,
    dz_base=dz_base,
    D=D_new,
    n_tubes=n_tubes_new,
    rho_a=rho_a,
    cp_a=cp_a,
    Lf=Lf,
    g=g,
    x_start=x_s0
)

baseline_Tchip_C = chip_temperature_from_pressure(P_task2)

print("\n================ BASELINE TASK 3 SYSTEM @ Ta,in = 25 C ================\n")
print(f"P_ev [kPa]              = {P_task2:.3f}")
print(f"Va [m/s]                = {Va_base:.3f}")
print(f"Tube length [m]         = {L_task3b:.6f}")
print(f"Outlet quality [-]      = {baseline_case['x_exit']:.6f}")
print(f"Outlet pressure [kPa]   = {baseline_case['P_exit_kPa']:.6f}")
print(f"Total heat transfer [W] = {baseline_case['q_total_W']:.6f}")
print(f"Predicted chip temp [C] = {baseline_Tchip_C:.6f}")
print(f"Fully condensed?        = {baseline_case['fully_condensed']}")


# ------------------------------------------------------------
```

```python
# SEARCH FOR FEASIBLE MODIFICATIONS (ranges can be changed)
# ------------------------------------------------------------
P_candidates = np.arange(180.0, 241.0, 5.0)      # kPa
Va_candidates = np.arange(6.5, 13.1, 0.5)        # m/s
L_candidates = np.arange(L_task3b, 0.801, 0.01) # m

solutions = []

for P_try in P_candidates:
   Tchip_try = chip_temperature_from_pressure(P_try)

   # chip must stay below 55 C
   if Tchip_try >= T_chip_target_C:
       continue

   for Va_try in Va_candidates:
       for L_try in L_candidates:
           sim = condenser_sim_variable_length(
               total_length_m=L_try,
               q_chip=q_chip,
               Ta_in_C=Ta_in_new_C,
               Va=Va_try,
               P_design=P_try,
               dz_base=dz_base,
               D=D_new,
               n_tubes=n_tubes_new,
               rho_a=rho_a,
               cp_a=cp_a,
               Lf=Lf,
               g=g,
               x_start=x_s0
           )

           # criterion: outlet saturated liquid
           if sim["x_exit"] <= 1e-4:
               # score = smallest overall deviation from baseline system
               score = (
                   abs(P_try - P_task2) / P_task2
                   + abs(Va_try - Va_base) / Va_base
                   + abs(L_try - L_task3b) / L_task3b
               )

               solutions.append({
                   "P_ev_kPa": P_try,
                   "Va_m_s": Va_try,
                   "Tube_length_m": L_try,
                   "Tube_length_cm": 100.0 * L_try,
                   "Outlet_pressure_kPa": sim["P_exit_kPa"],
                   "Outlet_quality": sim["x_exit"],
                   "Outlet_void_fraction": sim["alpha_exit"],
                   "Total_heat_transfer_W": sim["q_total_W"],
                   "Chip_temperature_C": Tchip_try,
                   "Fully_condensed": sim["fully_condensed"],
                   "Score": score
               })

# ------------------------------------------------------------
# RESULTS OF SEARCH
# ------------------------------------------------------------
if len(solutions) == 0:
   print("\nNo feasible modified design found in the search range.")
else:
   solutions_df = pd.DataFrame(solutions).sort_values("Score").reset_index(drop=True)

   print("\n================ FEASIBLE MODIFIED SYSTEMS ================\n")
   print(solutions_df.head(10).to_string(index=False))

   # choose best / least-modified feasible design
   best = solutions_df.iloc[0]

   # comparison table requested in Task 3(c)
   comparison_table = pd.DataFrame({
```

```
        "Quantity": [
            "Evaporator pressure [kPa]",
            "Air velocity [m/s]",
            "Tube length [m]",
            "Tube length [cm]",
            "Inlet air temperature [C]",
            "Outlet pressure [kPa]",
            "Outlet quality [-]",
            "Outlet void fraction [-]",
            "Total heat transfer to air [W]",
            "Predicted chip temperature [C]"
        ],
        "Task 3 system @ 25 C": [
            P_task2,
            Va_base,
            L_task3b,
            100.0 * L_task3b,
            Ta_in_new_C,
            baseline_case["P_exit_kPa"],
            baseline_case["x_exit"],
            baseline_case["alpha_exit"],
            baseline_case["q_total_W"],
            baseline_Tchip_C
        ],
        "Altered feasible system": [
            best["P_ev_kPa"],
            best["Va_m_s"],
            best["Tube_length_m"],
            best["Tube_length_cm"],
            Ta_in_new_C,
            best["Outlet_pressure_kPa"],
            best["Outlet_quality"],
            best["Outlet_void_fraction"],
            best["Total_heat_transfer_W"],
            best["Chip_temperature_C"]
        ]
    })

    print("\n================ TASK 3(c): COMPARISON TABLE ================\n")
    print(comparison_table.to_string(index=False))

    # save tables if needed
    solutions_df.to_csv("task3c_feasible_solutions.csv", index=False)
    comparison_table.to_csv("task3c_comparison_table.csv", index=False)
```

Output

```
================ BASELINE TASK 3 SYSTEM @ Ta,in = 25 C ================

P_ev [kPa]              = 285.630
Va [m/s]                = 6.500
Tube length [m]         = 0.205504
Outlet quality [-]      = 0.154904
Outlet pressure [kPa]   = 285.379219
Total heat transfer [W] = 286.992556
Predicted chip temp [C] = 56.202567
Fully condensed?        = False

=============== FEASIBLE MODIFIED SYSTEMS ===============

 P_ev_kPa  Va_m_s  Tube_length_m  Tube_length_cm  Outlet_pressure_kPa  Outlet_quality  Outlet_void_fraction  Total_heat_transfer_W   Chip_temperature_C  Fully_condensed   Score
270.00000 7.00000        0.23550        23.55040            269.75580         0.00000               0.00000              339.66000             53.98786             True 0.27763
275.00000 6.50000        0.25550        25.55040            274.69861         0.00000               0.00000              339.66000             54.70587             True 0.28052
270.00000 7.00000        0.24550        24.55040            269.75580         0.00000               0.00000              339.66000             53.98786             True 0.32629
275.00000 6.50000        0.26550        26.55040            274.69861         0.00000               0.00000              339.66000             54.70587             True 0.32918
275.00000 7.50000        0.23550        23.55040            274.74707         0.00000               0.00000              339.66000             54.70587             True 0.33704
270.00000 6.50000        0.26550        26.55040            269.68554         0.00000               0.00000              339.66000             53.98786             True 0.34669
275.00000 7.00000        0.25550        25.55040            274.70525         0.00000               0.00000              339.66000             54.70587             True 0.35744
275.00000 8.00000        0.22550        22.55040            274.75429         0.00000               0.00000              339.66000             54.70587             True 0.36531
270.00000 7.00000        0.25550        25.55040            269.75580         0.00000               0.00000              339.66000             53.98786             True 0.37495
275.00000 6.50000        0.27550        27.55040            274.69861         0.00000               0.00000              339.66000             54.70587             True 0.37784
```

```
================ TASK 3(c): COMPARISON TABLE ================

                      Quantity  Task 3 system @ 25 C  Altered feasible system
     Evaporator pressure [kPa]             285.63000                270.00000
            Air velocity [m/s]               6.50000                  7.00000
               Tube length [m]               0.20550                  0.23550
              Tube length [cm]              20.55040                 23.55040
     Inlet air temperature [C]              25.00000                 25.00000
         Outlet pressure [kPa]             285.37922                269.75580
            Outlet quality [-]               0.15490                  0.00000
      Outlet void fraction [-]               0.53969                  0.00000
  Total heat transfer to air [W]            286.99256                339.66000
  Predicted chip temperature [C]             56.20257                 53.98786
```

Appendix D - Task 4 Code

The code for Task 4 can be found below.

```
''' Task 4 '''

import numpy as np
import math
import matplotlib.pyplot as plt
from numpy.polynomial import polynomial as P

#define the five evaporator pressures (P2)
P2_vals = [160.0, 180.0, 200.0, 220.0, 240.0] #kPa

q_chip=340.0        # W
Ta_in_C=20.0        # C
Va=6.5              # m/s
dz=0.040          # m
D=0.003             # m
n_tubes=9           # -
rho_a=1.3           # kg/m^3
g=9.81               # m/s^2
segments_count = 20     #number of segments
A_chip = (6e-2)**2  #m^2
q_double_dot = q_chip/A_chip #W/m^2
T_design_chip = 55.0 + 273.15 #K
theta = 35 * np.pi/180.0 #radians

#from the image in the description of Task 2
x_s0= 0.999
alpha_s0 = 0.999
```

```
cp_a=1005.0         # J/kg·K
Lf=5.26e-3          # m

# Tsat(P2) constants
C0 = 20.472
C1 = -2579.9

#Lagrange interpolation for Thermophysical Properties

# ------------------------------
# R245fa saturation property data
# ------------------------------
T_data_C = np.array([15.1, 30.0, 46.0, 60.0])          #C

rho_l_data = np.array([1365.0, 1325.0, 1279.0, 1267.0])      # kg/m^3
rho_v_data = np.array([5.961, 10.169, 17.30, 25.68])         # kg/m^3
mu_l_data  = np.array([464e-6, 382e-6, 314e-6, 266e-6])      # Pa·s
mu_v_data  = np.array([9.93e-6, 10.4e-6, 11.0e-6, 11.5e-6])  # Pa·s
k_l_data   = np.array([0.0912, 0.0865, 0.0814, 0.0769])      # W/m·K
h_lv_data  = np.array([196.1e3, 187.0e3, 177.2e3, 168.0e3])  # J/kg0.
sigma_data = np.array([0.0153, 0.0134, 0.0114, 0.0096])      # N/m
cp_l_data  = np.array([1318.0, 1350.0, 1388.0, 1428.0])      # J/kg·K

def lagrange_interp(x, xdata, ydata):
    n = len(xdata)
    y = 0.0
    for i in range(n):
        Li = 1.0
        for j in range(n):
            if i != j:
                Li *= (x - xdata[j]) / (xdata[i] - xdata[j])
        y += ydata[i] * Li
    return y


def Tsat_from_P(P_kPa):
    P_Pa = P_kPa * 1000.0
    Tsat_K = C1 / (np.log(P_Pa) - C0)
    return Tsat_K


def r245fa_sat_props(P_kPa):
    Tsat_K = Tsat_from_P(P_kPa)
    Tsat_C = Tsat_K - 273.15

    rho_l = lagrange_interp(Tsat_C, T_data_C, rho_l_data)
    rho_v = lagrange_interp(Tsat_C, T_data_C, rho_v_data)
    mu_l  = lagrange_interp(Tsat_C, T_data_C, mu_l_data)
    mu_v  = lagrange_interp(Tsat_C, T_data_C, mu_v_data)
    k_l   = lagrange_interp(Tsat_C, T_data_C, k_l_data)
    h_lv  = lagrange_interp(Tsat_C, T_data_C, h_lv_data)
    sigma = lagrange_interp(Tsat_C, T_data_C, sigma_data)
    cp_l  = lagrange_interp(Tsat_C, T_data_C, cp_l_data)

    Pr_l = mu_l * cp_l / k_l

    return {
```

```
        "Tsat_K": Tsat_K,
        "Tsat_C": Tsat_C,
        "rho_l": rho_l,
        "rho_v": rho_v,
        "mu_l": mu_l,
        "mu_v": mu_v,
        "k_l": k_l,
        "h_lv": h_lv,
        "sigma": sigma,
        "cp_l": cp_l,
        "Pr_l": Pr_l
    }

units = {
    "Tsat_K": "K",
    "Tsat_C": "°C",
    "rho_l": "kg/m^3",
    "rho_v": "kg/m^3",
    "mu_l": "Pa·s",
    "mu_v": "Pa·s",
    "k_l": "W/m·K",
    "h_lv": "J/kg",
    "sigma": "N/m",
    "cp_l": "J/kg·K",
    "Pr_l": "-"
}


#repeated from Task 1 to define the segment calculating functio:
#Single Segment Condenser Element

def segment_model(
    q_chip=340.0,        # W
    Ta_in_C=20.0,        # °C
    Va=6.5,              # m/s
    P_in_kPa=220.0,      # kPa
    dz=0.040,            # m
    D=0.003,             # m
    n_tubes=9,           # -
    x_in=0.999,          # -
    rho_a=1.3,           # kg/m^3
    cp_a=1005.0,         # J/kg·K
    Lf=5.26e-3,          # m
    g=9.81               # m/s^2
):
    props = r245fa_sat_props(P_in_kPa)

    Tsat_K = props["Tsat_K"]
    rho_l = props["rho_l"]
    rho_v = props["rho_v"]
    mu_l = props["mu_l"]
    mu_v = props["mu_v"]
    k_l = props["k_l"]
    h_lv = props["h_lv"]
    sigma = props["sigma"]
    Pr_l = props["Pr_l"]
```

```
    Ta_in_K = Ta_in_C + 273.15

    # Mass flux
    A_passage = np.pi * D**2 / 4.0                      # m^2
    A_total = n_tubes * A_passage                       # m^2
    G = q_chip / (A_total * h_lv)                       # kg/m^2·s

    # Reynolds numbers
    Re_lo = G * D / mu_l
    Re_l = G * (1.0 - x_in) * D / mu_l
    Re_vo = G * D / mu_v

    # Martinelli parameter
    X_tt = ((1.0 - x_in) / x_in)**0.9 * (rho_v / rho_l)**0.5 * (mu_l / mu_v)**0.1

    # Traviss F_T
    if Re_l > 1125:
        F_T = 0.812 * (Pr_l**0.5) + 5.0 * np.log(1.0 + 5.0 * Pr_l) + 2.5 * np.log(0.0031 *
Re_l)
    elif Re_l > 50:
        F_T = 0.585 * (Pr_l**0.5) + 5.0 * np.log(1.0 + Pr_l * (0.0964 * Re_l**0.585 - 1.0))
    else:
        F_T = 0.5 * (Pr_l**0.5) * Re_l

    # Refrigerant-side heat transfer coefficient
    hr = (k_l / D) * 0.15 * (Pr_l**0.4) * (Re_l**0.9) * F_T * (1.0 / X_tt + 2.85 /
X_tt**0.476)

    # Inlet void fraction
    alpha_in = 1.0 / (
        1.0
        + ((1.0 - x_in) / x_in)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Segment UA and heat transfer
    Ar = n_tubes * np.pi * D * dz                        # m^2
    UA_air = 77.1 * (Va / 6.5)**0.5 * dz                # W/K
    UA_seg = 1.0 / (1.0 / (hr * Ar) + 1.0 / UA_air)     # W/K

    m_dot_a_seg = rho_a * Va * (2.0 * Lf) * dz          # kg/s
    C_a_seg = m_dot_a_seg * cp_a                        # W/K
    NTU = UA_seg / C_a_seg
    eps_seg = 1.0 - np.exp(-NTU)

    q_seg = C_a_seg * (Tsat_K - Ta_in_K) * eps_seg      # W

    # Exit quality
    x_out = x_in - q_seg / q_chip
    if x_out < 0.0:
        x_out = 1.0e-5
        q_seg = (x_in - x_out) * q_chip

    # Exit void fraction
    alpha_out = 1.0 / (
        1.0
```

```
        + ((1.0 - x_out) / x_out)**0.64
        * (rho_v / rho_l)**0.36
        * (mu_l / mu_v)**0.07
    )

    # Pressure drop model
    Gamma_in = G**2 * (
        x_in**2 / (rho_v * alpha_in)
        + (1.0 - x_in)**2 / (rho_l * (1.0 - alpha_in))
    )

    Gamma_out = G**2 * (
        x_out**2 / (rho_v * alpha_out)
        + (1.0 - x_out)**2 / (rho_l * (1.0 - alpha_out))
    )

    f_lo = 16.0 / Re_lo
    dPdz_lo = 2.0 * G**2 * f_lo / (D * rho_l)          # Pa/m

    if Re_vo < 2300.0:
        f_vo = 16.0 / Re_vo
    else:
        f_vo = 0.079 * Re_vo**(-0.25)

    rho_tp = 1.0 / (x_in / rho_v + (1.0 - x_in) / rho_l)
    Fr = G**2 / (rho_tp**2 * g * D)
    We = G**2 * D / (rho_tp * sigma)

    E = (1.0 - x_in)**2 + x_in**2 * (rho_l * f_vo) / (rho_v * f_lo)
    F = x_in**0.78 * (1.0 - x_in)**0.224
    H = (rho_l / rho_v)**0.91 * (mu_v / mu_l)**0.19 * (1.0 - mu_v / mu_l)**0.7

    phi_lo_sq = E + 3.24 * F * H / (Fr**0.045 * We**0.035)

    P_in_Pa = P_in_kPa * 1000.0
    P_out_Pa = P_in_Pa - phi_lo_sq * dPdz_lo * dz - (Gamma_out - Gamma_in)
    P_out_kPa = P_out_Pa / 1000.0

    return {
        "Tsat_C": props["Tsat_C"],          # °C
        "G": G,                             # kg/m^2·s
        "hr": hr,                           # W/m^2·K
        "UA_seg": UA_seg,                   # W/K
        "q_seg": q_seg,                     # W
        "x_out": x_out,                     # -
        "alpha_in": alpha_in,               # -
        "alpha_out": alpha_out,             # -
        "phi_lo_sq": phi_lo_sq,             # -
        "P_out_kPa": P_out_kPa              # kPa
    }

## Part 1
-------------------------------------------------------------------------------------------
------------------------------

#find T_sat and T_w
def T_wall_calculator(q_chip, A_chip, P_vals):
```

```
    '''
    Find T_w and the pressure that satisfies the chip temperature limit.
    '''
    T_chip_list = []
    for P_kPa in P_vals:
        props = r245fa_sat_props(P_kPa)
        Tsat  = props["Tsat_K"]
        rho_l = props["rho_l"]
        rho_v = props["rho_v"]
        k_l   = props["k_l"]
        Pr_l  = props["Pr_l"]
        sigma = props["sigma"]

        #to find T_w:

        Db = 0.0208 * theta * np.sqrt(sigma / (g * (rho_l - rho_v)))

        # B = (rho_v / rho_l)^0.581 * Pr_l^0.533
        # use the rearranged eq from notes: T_w =
[(1/207)(q_double_dot*Db/k_l)(k_l*T_sat/q_double_dot*Db)^0.745 *(B)^-1] + T_sat

        B = ((rho_v / rho_l)**0.581) * (Pr_l**0.533)
        T_wall = [(1/207)*(q_double_dot*Db/k_l)*((k_l*Tsat/q_double_dot*Db)**0.745)
*(B**(-1))] + Tsat

        T_chip_list.append(T_wall)

    return T_chip_list

#plot T_chip vs P_ev for Part 1 Deliverable
T_chip_list = T_wall_calculator(q_chip, A_chip, P2_vals)
T_chip_C = [T - 273.15 for T in T_chip_list]


## Use a curve-fit to find the evaporator pressure that satisfies the specified q_dot at
T_chip = 55C
P_array = np.array(P2_vals)
T_array = np.array(T_chip_list).flatten()

#fit a first degree polynomial to T_chip vs P_2 -> P_2(T_chip) since T_chip is independent
coeffs = np.polyfit(T_array - 273.15, P_array, deg=1)
poly   = np.poly1d(coeffs)

#evaluate P at T_chip = 55°C
P_design = poly(55.0)

#print(f"\n[Part 1] Design evaporator pressure: {P_design:.2f} kPa, chip will be at 55C")



## Part 2
-----------------------------------------------------------------------------------------------
------------------------

#part 2(a) Loop and Sim Setup for Full Condenser
def condenser_sim(q_chip, Ta_in_C, Va, P_design, dz, D, n_tubes, segments_count, rho_a,
cp_a, Lf, g):
```

```
    '''
    Iterates through the 20 condenser segments, returning exit states of all segments and
the total
    properties of heat transfer rate and pressure drop and final exit state of the
condenser exit.
    '''
    #at inlet of s0: saturated vapour at evaporator pressure enters condenser
    x = x_s0
    P_kPa = P_design

    records = []
    q_total   = 0.0
    dP_total  = 0.0

    for i in range(segments_count):
        res = segment_model(
            q_chip  = q_chip, Ta_in_C = Ta_in_C, Va = Va,
            P_in_kPa= P_kPa, dz = dz, D = D, n_tubes = n_tubes, x_in = x,
            rho_a = rho_a, cp_a = cp_a, Lf = Lf, g = g
            )

        x_out = res["x_out"]
        alpha_out = res["alpha_out"]
        P_out_kPa = res["P_out_kPa"]
        q_seg = res["q_seg"]

        records.append({
            "seg"      : i + 1,
            "x_out"    : x_out,
            "alpha_out": alpha_out,
            "P_out_kPa": P_out_kPa,
            "q_seg"    : q_seg
        })

        #total rate of heat transfer calc
        q_total  += q_seg
        #total pressure drop calc
        dP_total += (P_kPa - P_out_kPa)

        #next segment state information
        x = x_out
        P_kPa = P_out_kPa

    return records, q_total, dP_total, P_kPa, x, alpha_out


#Part 2(b) Find the total length of the condenser for x = 0 (qulaity to be equal to zero
exactly)
def best_length_search(q_chip, Ta_in_C, Va, P_design, dz, D, n_tubes,
                                rho_a, cp_a, Lf, g, segments_count):
    '''
    Finds the total condenser length for which x becomes zero.
    '''

    #initialize a total length and define properties
    L = 0.0
    P_kPa = P_design
```

```
    x = x_s0

    #set up the loop to search
    for seg in range(segments_count):
        if x <= 0.0001:
            break
        sols = segment_model(q_chip=q_chip, Ta_in_C=Ta_in_C, Va=Va,
                             P_in_kPa=P_kPa, dz=dz, D=D, n_tubes=n_tubes,
                             x_in=x, rho_a=rho_a, cp_a=cp_a, Lf=Lf, g=g
                             )
        L += dz
        P_kPa = sols["P_out_kPa"]
        x = sols["x_out"]

    return L

#export the total length that achieves a quality of zero by calling the function
L_x_0 = best_length_search(q_chip, Ta_in_C, Va,
                           P_design, dz, D, n_tubes,
                           rho_a, cp_a, Lf, g, segments_count)

#report the final length val for Part 2(b)
#print(f"\n[Part 2b] The total condenser length that achieves a quality of zero at the exit
is: {L_x_0:.3f} m" )

#-------------------------------------------------------------------------------------------
---------------------------
# TASK 4
#Part a - find the critical (maximum) heat flux
q_max_list = []
L_chip = 0.06 #chip length [m] (6 cm)

for P_kPa in P2_vals:
    properties = r245fa_sat_props(P_kPa)
    rho_l = properties["rho_l"]
    rho_v = properties["rho_v"]
    h_lv = properties["h_lv"]
    sigma = properties["sigma"]

    #find L_b, bubble length scale
    L_b = np.sqrt(sigma / (g * (rho_l - rho_v)))

    #find lambda_d, the Taylor instability wavelength
    lambda_D = 2 * np.pi * np.sqrt(3) * L_b

    #Zuber's critical heat flux formula
    q_max_Z = 0.131 * rho_v * h_lv * ((sigma * (rho_l - rho_v) * g) / (rho_v**2))**0.25

    #pick how to calculate q_max
    ratio = L_chip / L_b
    if ratio > 30:
        q_max_actual = 1.14 * q_max_Z
    elif 9 < ratio < 20:
        q_max_actual = (1.14 * lambda_D**2 / A_chip) * q_max_Z
    else:
        q_max_actual = q_max_Z  #if between ranges, use the theoretical max value
```

```
    #convert to Watts
    q_max_list.append(q_max_actual * A_chip)

#Plot to compare
plt.figure()
plt.plot(P2_vals, q_max_list, 'b-o', label='Critical Heat Flux (CHF)[W]')
plt.axhline(q_chip, color='k', linestyle='--', label='Design Load (340W)')
plt.xlabel('Evaporator Pressure [kPa]')
plt.ylabel('Power [W]')
plt.title('CHF [W] vs Evaporator Pressure [KPa]')
plt.legend()
plt.show()

#-------------------------------------------------------------------------------------------
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#Task 4 Part b - find the chip temperature at the CHF and modify the condenser design to
optimize cooling at this higher bound

#find the CHF value for P_design
properties = r245fa_sat_props(P_design)
rho_l = properties["rho_l"]
rho_v = properties["rho_v"]
h_lv = properties["h_lv"]
sigma = properties["sigma"]

#find L_b, bubble length scale
L_b = np.sqrt(sigma / (g * (rho_l - rho_v)))

#find lambda_d, the Taylor instability wavelength
lambda_D = 2 * np.pi * np.sqrt(3) * L_b

#Zuber's critical heat flux formula
q_max_Z = 0.131 * rho_v * h_lv * ((sigma * (rho_l - rho_v) * g) / (rho_v**2))**0.25

#pick how to calculate q_max
ratio = L_chip / L_b
if ratio > 30:
    q_max_actual = 1.14 * q_max_Z
elif 9 < ratio < 20:
    q_max_actual = (1.14 * lambda_D**2 / A_chip) * q_max_Z
else:
    q_max_actual = q_max_Z  #if between ranges, use the theoretical max value

#convert to Watts
q_CHF_design = q_max_actual * A_chip

print("The CHF corresponding to the Design Pressure (W):" , f"{q_CHF_design:.3f}" )

#find the chip temp at the CHF limit by providing the max heat transfer rate, the area of
the heat transfer, and the design chip temp, and design pressure
T_chip_max_boundary = T_wall_calculator(q_CHF_design, A_chip, [P_design])[0]
print(f"Chip Temperature at CHF [C] : {T_chip_max_boundary - 273.15}")

#-----------------------------------------------------
#condenser improvement and redesign simulations
#-----------------------------------------------------
```

```
cases = [1, 2, 3, 4]
for case in cases:
  if case == 1:
    Va_new = Va * 1.9
    rec_c1, q_tot_c1, dP_c1, Pf_c1, xf_c1, af_c1 = condenser_sim(q_CHF_design, Ta_in_C,
Va_new, P_design,
                                                        dz, D, n_tubes, segments_count,
                                                        rho_a, cp_a, Lf, g)


    print(f"\n[Part b] Case {case} Condenser Simulation Results")
    print(f"{'Seg':>4} {'x_out':>8} {'α_out':>8} {'P_out (kPa)':>10} {'q_seg (W)':>10}")

    for r in rec_c1:
        print(f"{r['seg']:>4} {r['x_out']:>8.5f} {r['alpha_out']:>8.5f}
{r['P_out_kPa']:>12.4f} {r['q_seg']:>10.4f}")

    print(f"\nTotal refrigerant pressure drop: {dP_c1:.4f} kPa")
    print(f"Total condenser heat transfer: {q_tot_c1:.2f} W")
    print(f"Condenser exit pressure: {Pf_c1:.4f} kPa")
    print(f"Condenser exit quality: {xf_c1:.6f}")

    if xf_c1 <= 0.0001:
        print("Exit state: saturated liquid -> full condensation achieved")
    else:
        print("Exit state: two-phase mixture")

  elif case == 2:
    P_new = P_design * 1.3
    n_tubes_new = n_tubes + 3
    rec_c2, q_tot_c2, dP_c2, Pf_c2, xf_c2, af_c2 = condenser_sim(q_CHF_design, Ta_in_C, Va,
P_new,
                                                        dz, D, n_tubes_new, segments_count,
                                                        rho_a, cp_a, Lf, g)
    print(f"\n[Part b] Case {case} Condenser Simulation Results")

    print(f"{'Seg':>4} {'x_out':>8} {'α_out':>8} {'P_out (kPa)':>10} {'q_seg (W)':>10}")

    for r in rec_c2:
        print(f"{r['seg']:>4} {r['x_out']:>8.5f} {r['alpha_out']:>8.5f}
{r['P_out_kPa']:>12.4f} {r['q_seg']:>10.4f}")

    print(f"\nTotal refrigerant pressure drop: {dP_c2:.4f} kPa")
    print(f"Total condenser heat transfer: {q_tot_c2:.2f} W")
    print(f"Condenser exit pressure: {Pf_c2:.4f} kPa")
    print(f"Condenser exit quality: {xf_c2:.6f}")

    if xf_c2 <= 0.0001:
        print("Exit state: saturated liquid -> full condensation achieved")
    else:
        print("Exit state: two-phase mixture")

  elif case == 3:
    segments_count_new = segments_count + 10

    rec_c3, q_tot_c3, dP_c3, Pf_c3, xf_c3, af_c3 = condenser_sim(q_CHF_design, Ta_in_C, Va,
P_design,
```

```
                                                        dz, D, n_tubes, segments_count_new,
                                                        rho_a, cp_a, Lf, g)
    print(f"\n[Part b] Case {case} Condenser Simulation Results")

    print(f"{'Seg':>4} {'x_out':>8} {'α_out':>8} {'P_out (kPa)':>10} {'q_seg (W)':>10}")

    for r in rec_c3:
        print(f"{r['seg']:>4} {r['x_out']:>8.5f} {r['alpha_out']:>8.5f}
{r['P_out_kPa']:>12.4f} {r['q_seg']:>10.4f}")

    print(f"\nTotal refrigerant pressure drop: {dP_c3:.4f} kPa")
    print(f"Total condenser heat transfer: {q_tot_c3:.2f} W")
    print(f"Condenser exit pressure: {Pf_c3:.4f} kPa")
    print(f"Condenser exit quality: {xf_c3:.6f}")

    if xf_c3 <= 0.0001:
        print("Exit state: saturated liquid -> full condensation achieved")
    else:
        print("Exit state: two-phase mixture")

  else:

    P_new = P_design * 1.3
    n_tubes_new = n_tubes + 5
    segments_count_new = segments_count + 4 #24 segments

    rec_c4, q_tot_c4, dP_c4, Pf_c4, xf_c4, af_c4 = condenser_sim(q_CHF_design, Ta_in_C, Va,
P_new,
                                                        dz, D, n_tubes_new,
segments_count_new,
                                                        rho_a, cp_a, Lf, g)


    print(f"\n[Part b] Case {case} Condenser Simulation Results")
    print(f"{'Seg':>4} {'x_out':>8} {'α_out':>8} {'P_out (kPa)':>10} {'q_seg (W)':>10}")

    for r in rec_c4:
        print(f"{r['seg']:>4} {r['x_out']:>8.5f} {r['alpha_out']:>8.5f}
{r['P_out_kPa']:>12.4f} {r['q_seg']:>10.4f}")

    print(f"\nTotal refrigerant pressure drop: {dP_c4:.4f} kPa")
    print(f"Total condenser heat transfer: {q_tot_c4:.2f} W")
    print(f"Condenser exit pressure: {Pf_c4:.4f} kPa")
    print(f"Condenser exit quality: {xf_c4:.6f}")

    if xf_c4 <= 0.0001:
        print("Exit state: saturated liquid -> full condensation achieved")
    else:
        print("Exit state: two-phase mixture")

# ------------------------------------------------------------------
#Task 4 Part c
##find the minimum V_a required to maintain T_chip under 65 C while chip heat output
fluctuates between 300-650W

q_range = np.arange(300, 651, 50) #[W], take steps of size 50W for exploring the entire
range
```

```
Va_min = []

print(f"\n{'Heat Load [W]':<14} {'Max Allowable P [kPa]':<24} {'Min Air Velocity
[m/s]':<24}")

#loop through the fluctuating chip heat output values to find V_a min that satisfies T_chip
< 65
for q in q_range:
    #find the pressure value for which T_chip = 65 C by checking where the pressure curve
touches 65C
    P_axis = np.linspace(100, 500, 100) #kPa

    #find T_chip in [degrees C] at each q
    T_test_result_array = np.array(T_wall_calculator(q, A_chip, P_axis)).flatten() - 273.15

    #interpolate to find the pressure corresponding to T=65
    P_limit = np.interp(65.0, T_test_result_array, P_axis)

    #define a test starting value for the air velocity to be looped
    va_test = 0.5
    va_found = False

    #find the min air velocity required at P_limit to achieve saturated liquid (i.e. exit
quality = 0)
    while va_test <= 20.0:
        rec_min, q_tot_min, dP_min, Pf_min, xf_min, af_min= condenser_sim(q, Ta_in_C,
va_test, P_limit, dz, D, (n_tubes+5), (segments_count + 4)
                                                , rho_a, cp_a, Lf, g)
        if xf_min <= 0.0001:
            Va_min.append(va_test)
            print(f"{q:<15.2f} {P_limit:<25.2f} {va_test:<25.3f}")
            va_found = True
            break

        va_test += 0.1 #increase velocity for next loop

    if not va_found:
        print(f"{q:<15.2f} {P_limit:<25.2f} {'FAILED':<25}")

    # check performance metrics of the condenser
    #print (rec_min, q_tot_min, dP_min, Pf_min, xf_min, af_min)

def visualize_interpolation(q_target):
  P_axis = np.linspace(100, 500, 100)

  T_results = np.array(T_wall_calculator(q_target, A_chip, P_axis)).flatten() - 273.15

  P_limit = np.interp(65.0, T_results, P_axis)

  plt.figure(figsize=(8, 6))

  #plot the curve for the chip
  plt.plot(P_axis, T_results, 'b-', label=f'Boiling Curve at {q_target}W')
  plt.axhline(y=65.0, color='r', linestyle='--', label=f'Temperature Limit of 65 C')
  plt.axvline(x=P_limit, linestyle=':', label=f'Pressure at the Limit: {P_limit:.2f} kPa')

  plt.xlabel('Evaporator Pressure [kPa]')
```

```
  plt.ylabel('Chip Temperature [C]')
  plt.grid(True, alpha=0.3)
  plt.legend()
  plt.show()

#plot the interpolation for maxium q
visualize_interpolation(650.0)

#plot the minimum air velocity requried to satisfy 65C vs the varying heat load
plt.figure(figsize=(8, 6))
plt.plot(q_range, Va_min, '-o', label='Minimum Required Air Velocity [m/s]')
plt.xlabel('Heat Load [W]')
plt.ylabel('Minimum Air Velocity [m/s]')
plt.grid(True, alpha=0.3)
plt.legend()
plt.show()
```